

AARTI STEEL INTERNATIONAL LIMITED
Ludhiana, Punjab

INDUSTRIAL INTERNSHIP REPORT

Analysis of Macrostructural Defects in Rolled Steel Products

with Detailed Metallurgical Investigation of Vanadium Micro-alloyed Steel (Heat No. 1907)



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Chapter 1

Introduction

Chapter 1: Introduction

1.1 Background

Rolled steel products such as bars and billets are widely used in the automotive, construction, and general engineering industries. The quality and reliability of these products depend strongly on the soundness of the internal and surface macrostructure, which is established during continuous casting and subsequently inherited through the rolling process. Any discontinuity introduced during solidification such as segregation, porosity, or entrapped non-metallic material can persist through hot rolling and manifest as a macrostructural defect in the final rolled product.

Macrostructural defects are of particular concern because, even when small in size, they act as stress concentration sites and can significantly reduce the mechanical performance, fatigue life, and machinability of the finished component. Timely identification and characterization of such defects is therefore essential for maintaining product quality and for tracing the defect back to its metallurgical origin in the casting process.

1.2 Importance of Macrostructural Defect Characterization

- Prevents premature failure of critical automotive and structural components.
- Enables root-cause identification casting related vs. rolling related defects.
- Supports corrective action in continuous casting parameters (superheat, casting speed, electromagnetic stirring, soft reduction).
- Improves overall yield and reduces rejection at the customer end.

1.3 Need for the Present Investigation

During routine inspection of rolled bar products, surface-breaking defects were observed on certain samples. A systematic investigation was therefore undertaken in two stages. In the first stage, defective samples of Heat No. A1726 (Grade 080A47) were examined using macro-etch testing and Positive Material Identification (PMI) to check whether the defect region showed any measurable difference in chemical composition compared to sound material, both before and after surface grinding. In the second stage, a more detailed metallurgical investigation was carried out on a Vanadium micro-alloyed steel, Heat No. 1907, using metallographic sample preparation, stereo and optical microscopy, etching, and cross-sectional crack analysis to understand the morphology, depth, and probable origin of the surface defect.

1.4 Objectives

- Study the macrostructural defects occurring in rolled steel products.
- Understand the origin and mechanism of defect formation during continuous casting.
- Perform PMI (XRF) analysis on defect and non-defect regions.
- Compare elemental composition before and after surface grinding.
- Correlate the observed defects with the continuous casting process.
- Carry out detailed metallographic characterization of a surface defect in Vanadium micro-alloyed steel (Heat No. 1907), including crack length, crack depth, crack spacing, inclusion assessment, and grain size determination.

Chapter 2

Preliminary Investigation and Analysis on Heat No. A1726

Chapter 2: Preliminary Investigation and Analysis on Heat No. A1726

This chapter presents the first stage of the investigation, carried out on defective rolled bar samples of Heat No. A1726, Grade 080A47.

2.1 Sample Details

Parameter	Details
Heat No.	A1726
Grade	080A47
Product	Rolled Bar
Dimension	58 mm diameter
Type of Defects Observed	Centerline Segregation, Sub-Surface Crack
No. of Defected Samples Observed	3

Grade 080A47 is a medium carbon steel (~0.45% C) offering good strength, toughness, machinability, and heat-treatment response. It is widely used for shafts, axles, gears, and other automotive components requiring high strength and wear resistance.

2.2 Types of Macrostructural Defects Observed in Rolled Steel

Before analysing the samples in detail, the common macrostructural defects encountered in rolled steel products were reviewed, along with their typical causes and effects.

2.2.1 Centerline Segregation

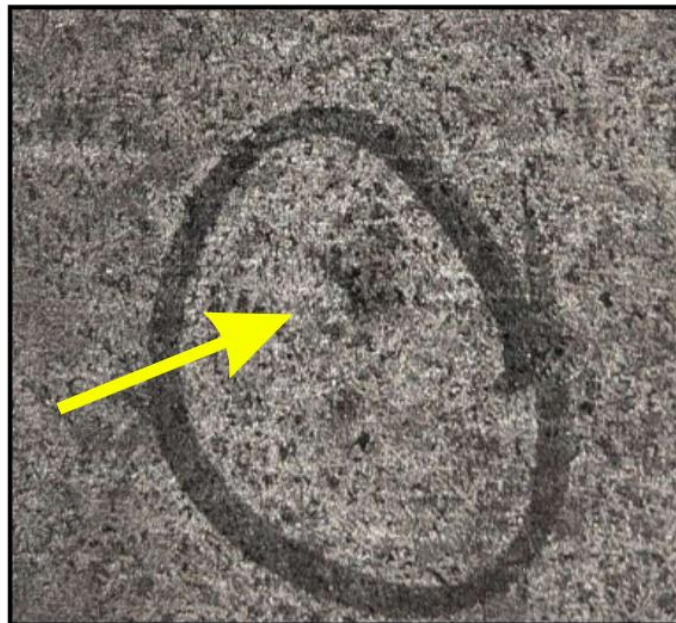


Figure 2.1: Macro-etched samples showing Centerline Segregation.

Central segregation refers to the localized enrichment of alloying and impurity elements principally carbon, phosphorus, sulfur, and manganese at the centerline of the billet. This phenomenon arises from solute redistribution during the final stage of solidification, wherein the last liquid to solidify, being progressively rejected of solute by the surrounding dendritic structure, becomes enriched and concentrates at the thermal center of the cast section.

Several process-related factors contribute to the severity of central segregation. High superheat delays the onset of solidification and promotes a deeper liquid pool, both of which extend the time available for solute-rich liquid to migrate toward the center before final solidification. High casting speeds compound this effect by increasing the depth of the liquid pool further and reducing the effectiveness of heat extraction. Inadequate electromagnetic stirring (EMS) fails to homogenize the melt or refine the solidifying structure, allowing solute-rich liquid to accumulate rather than being redistributed. Similarly, the absence of soft reduction at the final stage of solidification means that the natural shrinkage occurring as the strand solidifies is not compensated for, allowing solute-enriched liquid to be drawn toward the centerline rather than being displaced outward.

The consequences of central segregation are significant from both a metallurgical and operational standpoint. The enriched region typically exhibits reduced ductility, as the concentrated impurities disrupt the uniformity of the matrix. This segregation is often visibly manifested as a banded microstructure upon etching, reflecting the compositional inhomogeneity along the centerline. In terms of downstream processing, the segregated zone tends to result in poor machinability, since the altered local chemistry can cause inconsistent tool wear and surface finish. Perhaps most critically, central segregation increases the tendency for center cracking, as the compositional and structural discontinuities act as preferential sites for crack initiation and propagation during subsequent thermal or mechanical processing.

2.2.2 Center Porosity

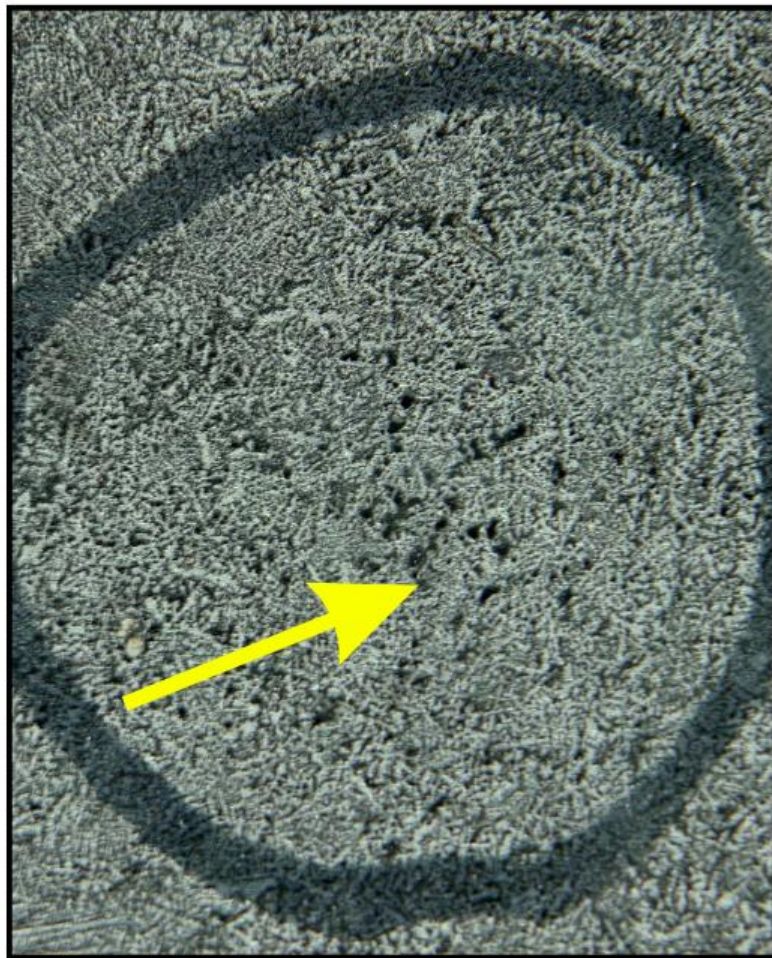


Figure 2.2: Macro-etched samples showing Center porosity.

Centerline porosity refers to the formation of small shrinkage cavities at the billet center, occurring when insufficient liquid metal is available to feed the contracting solidification front during the final stage of casting. As the strand solidifies, the volumetric shrinkage inherent to the solidification process must be continuously compensated for by the inward flow of liquid metal; when this feeding is inadequate, minute voids form at the thermal center where solidification concludes last.

This defect is driven by several interrelated factors. Solidification shrinkage itself is an unavoidable metallurgical phenomenon, but its effects are magnified when the mushy zone the semi-solid region between the liquidus and solidus fronts is deep, as this restricts the ability of liquid metal to penetrate and feed the shrinking core. Improper feeding practices, whether due to casting parameters or equipment limitations, further exacerbate the problem, as does the absence of soft reduction, which would otherwise mechanically compensate for shrinkage by gently compressing the strand at the appropriate stage of solidification.

The resulting internal voids compromise the integrity of the billet in multiple ways. They directly contribute to reduced tensile strength, since the porosity effectively reduces the load-bearing cross-section and introduces stress concentrators within the material. Billets affected by centerline porosity are also prone to ultrasonic testing (UT) rejection, as these internal discontinuities are readily detected by standard quality inspection methods. Furthermore, the voids themselves can serve as crack initiation sites during subsequent thermal or mechanical processing, posing a risk to the structural performance of the final product.

2.2.3 Pinholes

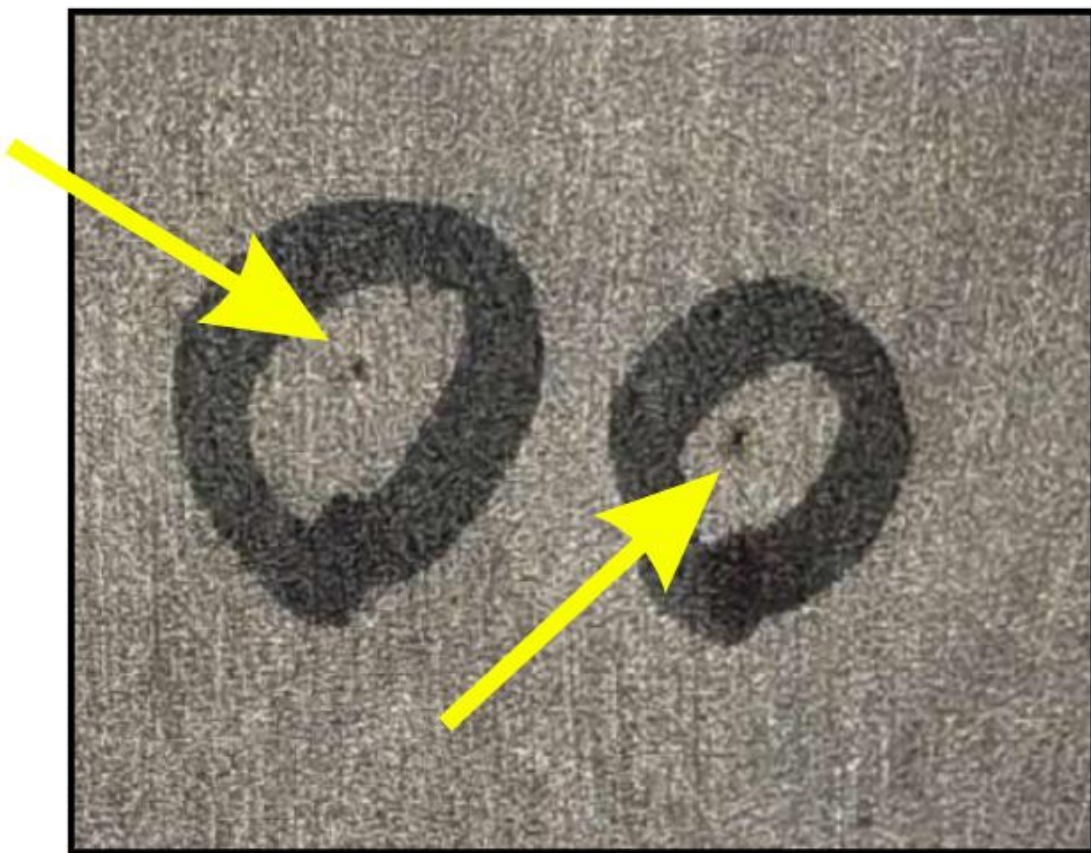


Figure 2.3: Macro-etched samples showing Center Pinhole.

Pinholes are small, spherical gas cavities that become trapped within the steel matrix during solidification. Unlike shrinkage-related defects, pinholes originate from gas evolution or entrapment rather than a lack of liquid feeding, giving them a characteristic rounded morphology.

The formation of pinholes is typically attributed to hydrogen pickup during steelmaking or casting, where dissolved hydrogen precipitates out as gas bubbles upon solidification. Carbon monoxide evolution, resulting from reactions between dissolved carbon and oxygen, is another common source of gas entrapment. Reoxidation of the melt, often occurring due to inadequate protection of the steel stream from atmospheric exposure, can introduce additional oxygen that reacts to form gas pockets. Underlying many of these causes is poor deoxidation practice, which leaves residual dissolved gases in the melt that are not adequately removed prior to casting.

The presence of pinholes has notable consequences for product quality. Once the billet undergoes rolling, these subsurface gas cavities often manifest as surface defects, compromising the visual and functional quality of the finished product. Pinholes also contribute to poor fatigue life, as the cavities act as stress risers that promote fatigue crack initiation under cyclic loading. Collectively, these effects result in reduced product quality and may necessitate downgrading or rejection of affected material.

2.2.4 Subsurface Crack



Figure 2.4: Macro-etched samples showing Subsurface Crack.

A subsurface crack is an internal crack located just below the billet surface, forming as a result of localized stress concentration. Such stress concentrations are commonly induced by chemical segregation, the presence of non-metallic inclusions, or thermal stresses generated during casting, all of which create a locally weakened region that is prone to cracking either during the casting process itself or in subsequent rolling operations.

Several process conditions contribute to the formation of subsurface cracks. Excessive secondary cooling can generate steep thermal gradients that induce high internal stresses, while thermal stress in general arising from uneven cooling rates across the cross-section further increases susceptibility to cracking. Corner overcooling, a common issue in billet casting, creates localized regions of accelerated cooling that are particularly prone to stress concentration. Additionally, the segregation and inclusion concentration discussed in earlier sections can compound these thermal effects by introducing chemically or physically weakened zones that serve as preferential crack initiation points.

The effects of subsurface cracking are considerable, as these internal flaws often propagate during rolling, resulting in visible rolling defects on the finished product. If not caught early, crack propagation can continue under the stresses imposed by further processing or service loading, ultimately leading to product rejection at quality control stages. More broadly, the presence of subsurface cracks represents a reduction in the structural integrity of the material, raising concerns about its performance in demanding applications.

2.2.5 Refractory Defect

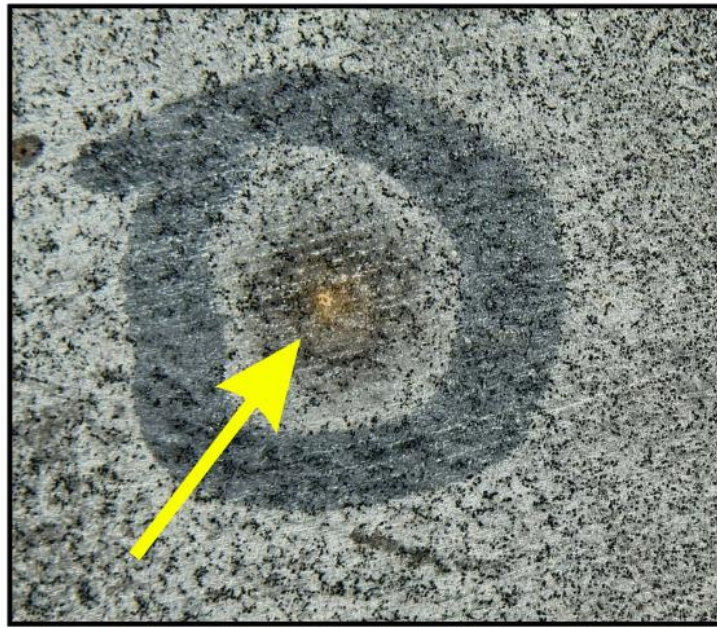


Figure 2.5: Macro-etched samples showing Refractory Defect.

A refractory defect arises from the entrainment of refractory particles originating from ladles, tundishes, nozzles, or other refractory linings into the molten steel during processing. These non-metallic particles, once trapped within the solidifying steel, act as localized stress concentration sites, contributing to overall quality deterioration and creating potential locations for crack initiation during subsequent rolling or in-service loading.

The introduction of refractory particles into the melt is generally attributable to refractory erosion, whereby the physical wear of refractory linings under the flow and thermal exposure of molten steel releases particulate matter into the stream. Thermal shock cracking of refractory materials, caused by rapid or repeated temperature fluctuations, can similarly generate fragments that become entrained in the steel. Turbulent steel flow within the ladle, tundish, or during pouring increases the likelihood of particle entrainment by promoting greater interaction between the steel and refractory surfaces. This is often compounded by nozzle and tundish wear, as well as by the use of poor refractory quality materials that are inherently more susceptible to degradation under casting conditions.

The consequences of refractory defects are primarily related to the introduction of non-metallic inclusions into the steel matrix, which act as discontinuities within an otherwise homogeneous structure. These inclusions frequently give rise to surface and subsurface defects that are detectable during quality inspection. As with other inclusion-related issues, refractory defects also promote crack initiation, particularly under the stress conditions encountered during rolling. Ultimately, the presence of these defects results in reduced mechanical properties, undermining the reliability and performance of the finished steel product.

2.3 Macro-Etch Testing

- A metallurgical examination technique used to reveal the macrostructure of steel and detect internal defects.
- The polished sample surface was treated with a suitable etchant (50% diluted hydrochloric acid) to highlight structural features.
- The test helps identify defects such as segregation, porosity, blowholes, shrinkage cavities, cracks and inclusions, and provides a visual assessment of material quality and solidification pattern.

- In this study, macro-etching was performed on samples containing surface defects to investigate their origin. The etched surface was examined for abnormal structural features and segregation patterns near the defect region.
- The results helped determine whether the observed defects were associated with solidification-related discontinuities or compositional segregation.
- Macro-etching is a simple, economical, and effective method for preliminary defect investigation.
- The test was conducted in accordance with ASTM E381 – Standard Method for Macroetch Testing of Steel Bars, Billets, Blooms, and Forgings.

2.4 Positive Material Identification (PMI)

- Hand-held XRF (X-Ray Fluorescence) is a non-destructive technique used for rapid elemental analysis of metals.
- The instrument irradiates the sample with X-rays, causing the elements present to emit characteristic fluorescent X-rays, which are analyzed to determine elemental composition.
- Analysis was performed at both defect and non-defect locations to identify compositional variations, with multiple readings (sparks) taken at each location to improve reliability.
- Elements analyzed: Mn, Cr, Ni, Mo, Si and Cu.
- Limitation: Carbon cannot be accurately detected by XRF due to its low atomic number.



Figure 2.6: Hand-held XRF (PMI) instrument used for elemental analysis

2.5 Ladle Chemistry of Grade 080A47

Cu	V	Al	C	Mn	S	P	Si	Cr	Ni	Mo
0.068	0.0026	0.023	0.485	0.90	0.015	0.018	0.223	0.334	0.078	0.013

Table 2.1: Ladle chemistry (weight %) of Heat No. A1726, Grade 080A47

2.6 Experiment 1: Linear PMI Analysis

PMI readings were taken along a line from the defect toward the sample center and beyond, using the following spark positions:

- P1 → Defect
- P2 → Between defect and center
- P3 → Center
- P4 → Opposite mid-radius

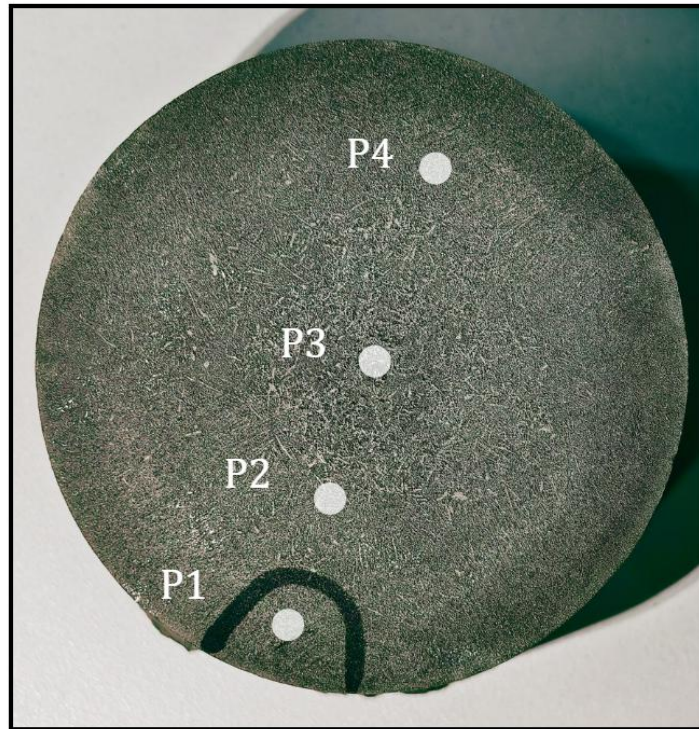


Figure 2.7: PMI spark locations – Experiment 1 (Linear PMI Analysis)

Spark Location	Sample 1 Mn (%)	Sample 1 Cr (%)	Sample 2 Mn (%)	Sample 2 Cr (%)	Sample 3 Mn (%)	Sample 3 Cr (%)
P1 – Defect	0.920	0.299	0.910	0.290	0.950	0.267
P2 – Between defect & center	0.890	0.310	0.910	0.309	0.910	0.320
P3 – Center	0.960	0.308	0.860	0.342	0.970	0.305
P4 – Opposite mid-radius	0.900	0.312	0.960	0.303	0.870	0.265

Table 2.2: Mn and Cr content (%) at spark locations P1–P4, Experiment 1 (all three samples)

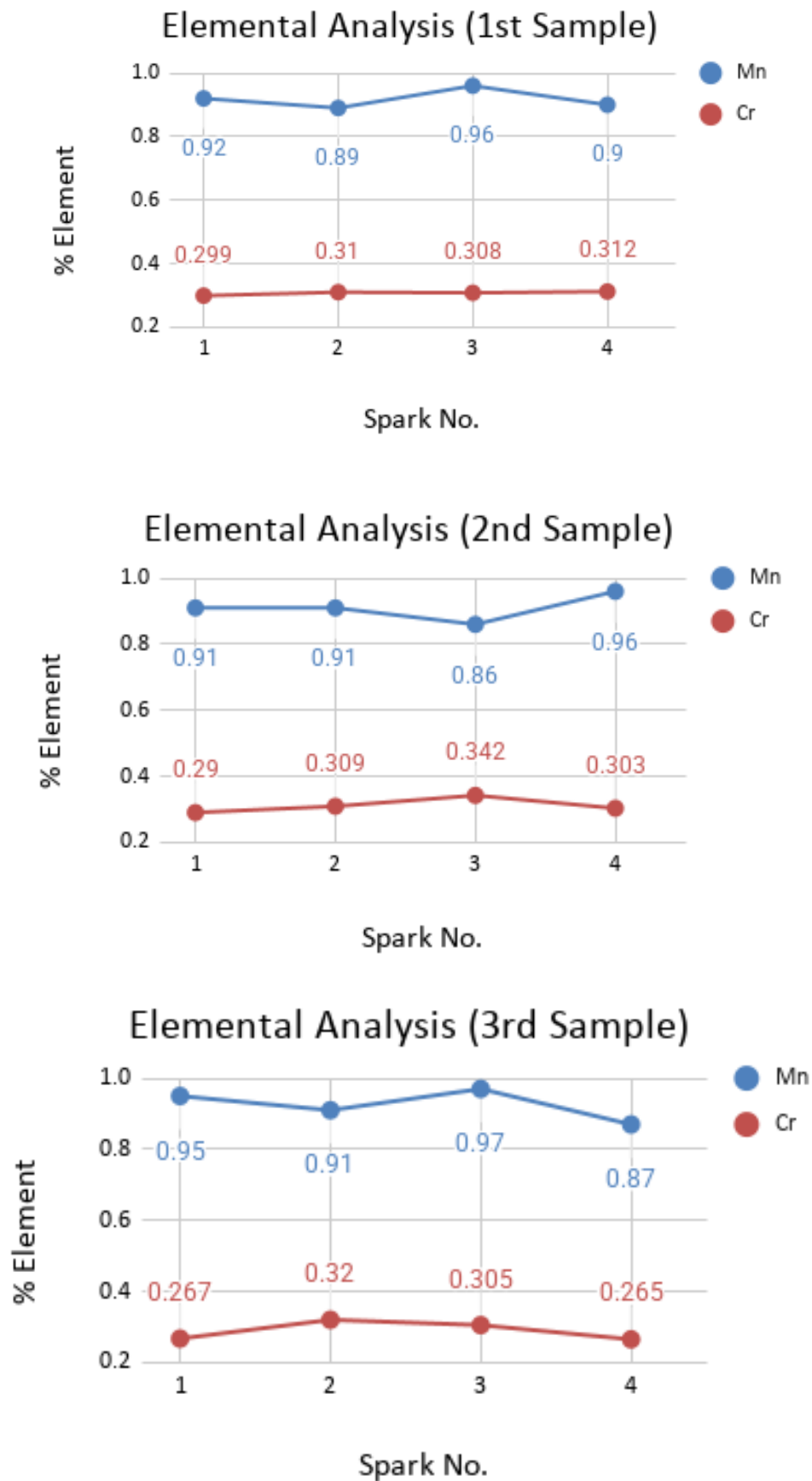
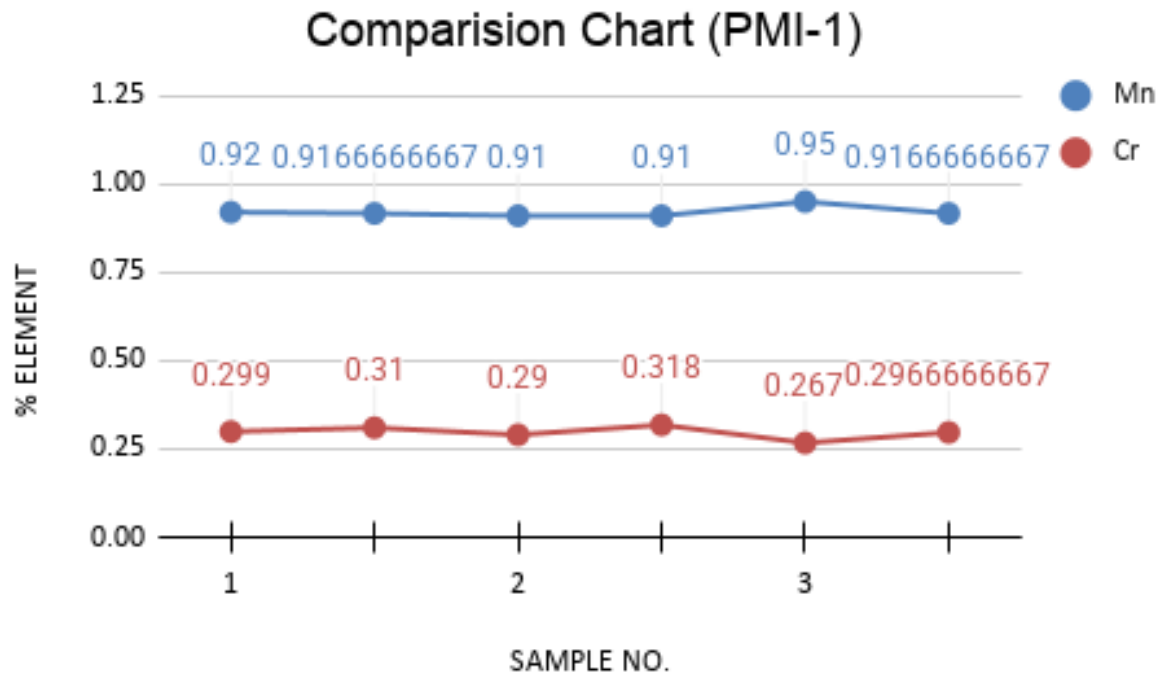


Figure 2.8: Mn and Cr variation across spark locations – Experiment 1

Sample No.	Defect (P1) Mn (%)	Non-defect Avg. Mn (%)	Defect (P1) Cr (%)	Non-defect Avg. Cr (%)
1	0.920	0.9167	0.299	0.3100
2	0.910	0.9100	0.290	0.3180
3	0.950	0.9167	0.267	0.2967

Table 2.3: Comparison of defect-location composition vs. average non-defect composition – Experiment 1



Observations

- A slight increase in Mn content was observed at the 3rd spark location (center) in two out of three samples.
- The center location consistently exhibited higher Mn values compared to the adjacent measurement points.
- This trend suggests a possible enrichment of Mn at the center region, which may be associated with centerline segregation during solidification.
- Due to the relatively higher Mn value at the center, the average non-defect composition becomes closer to the defect composition, reducing the apparent difference between them.
- Cr values did not show a similar consistent trend and remained relatively uniform across all measurement locations.

2.7 Experiment 2: PMI Analysis on Macro-etched Surface

A radial arrangement of spark positions around the defect was used on the macro-etched surface:

- P1 → Defect
- P2 → Opposite mid-radius
- P3 → Right mid-radius (to P1)
- P4 → Left mid-radius (to P1)

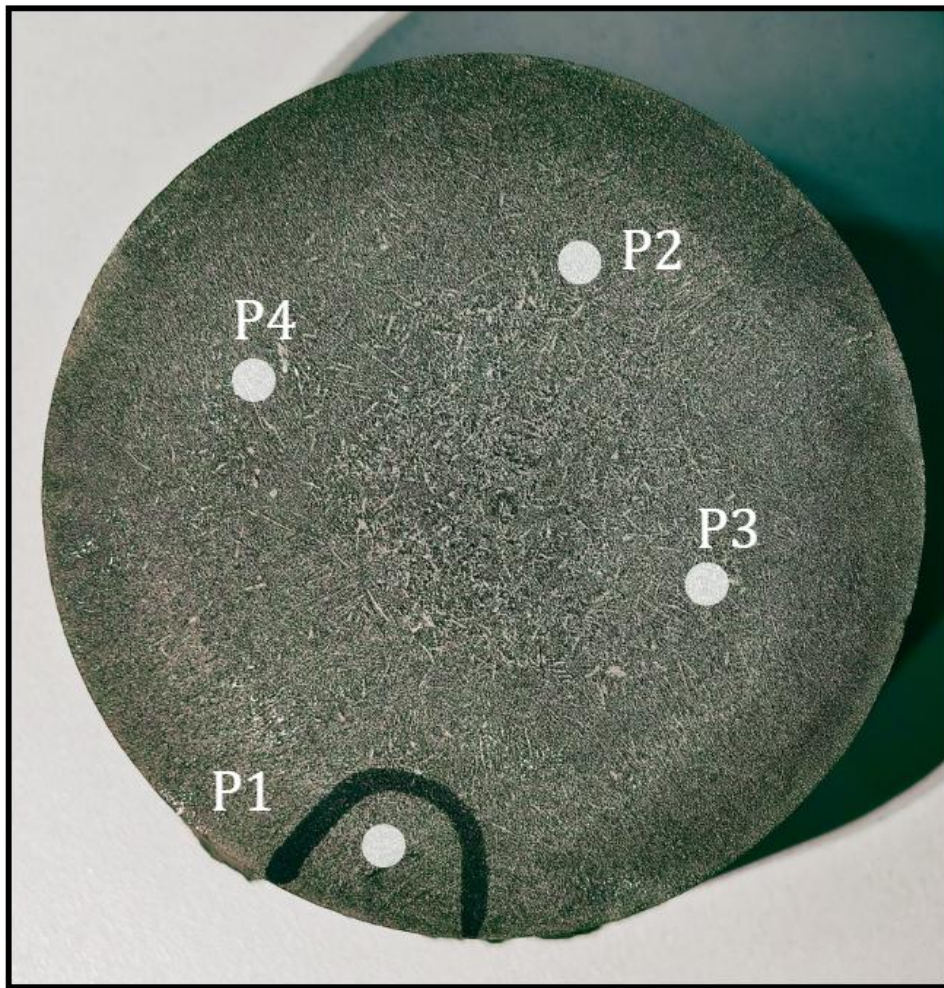


Figure 2.10: PMI spark locations – Experiment 2 (Macro-etched surface)

Spark Location	Sample 1 Mn (%)	Sample 1 Cr (%)	Sample 2 Mn (%)	Sample 2 Cr (%)	Sample 3 Mn (%)	Sample 3 Cr (%)
P1 – Defect	0.880	0.308	0.920	0.299	0.970	0.283
P2 – Opposite mid-radius	0.930	0.319	0.870	0.351	0.990	0.300
P3 – Right mid-radius	0.920	0.318	0.840	0.313	0.880	0.322
P4 – Left mid-radius	0.900	0.325	0.960	0.269	0.940	0.278

Table 2.4: Mn and Cr content (%) at spark locations P1–P4, Experiment 2 (all three samples)

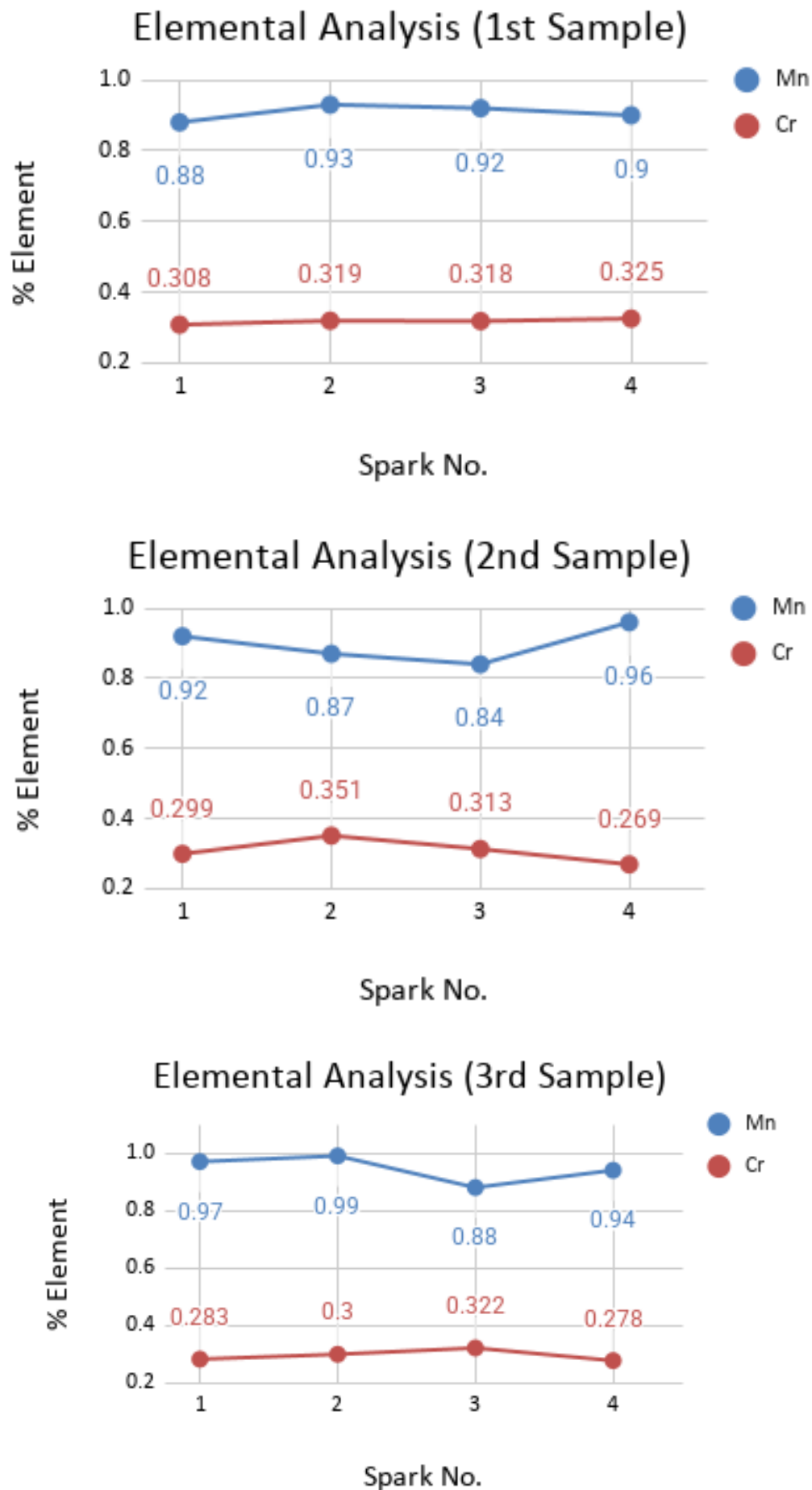


Figure 2.11: Mn and Cr variation across spark locations – Experiment 2

Sample No.	Defect (P1) Mn (%)	Non-defect Avg. Mn (%)	Defect (P1) Cr (%)	Non-defect Avg. Cr (%)
1	0.880	0.9167	0.308	0.3207
2	0.920	0.8900	0.299	0.3110
3	0.970	0.9367	0.283	0.3000

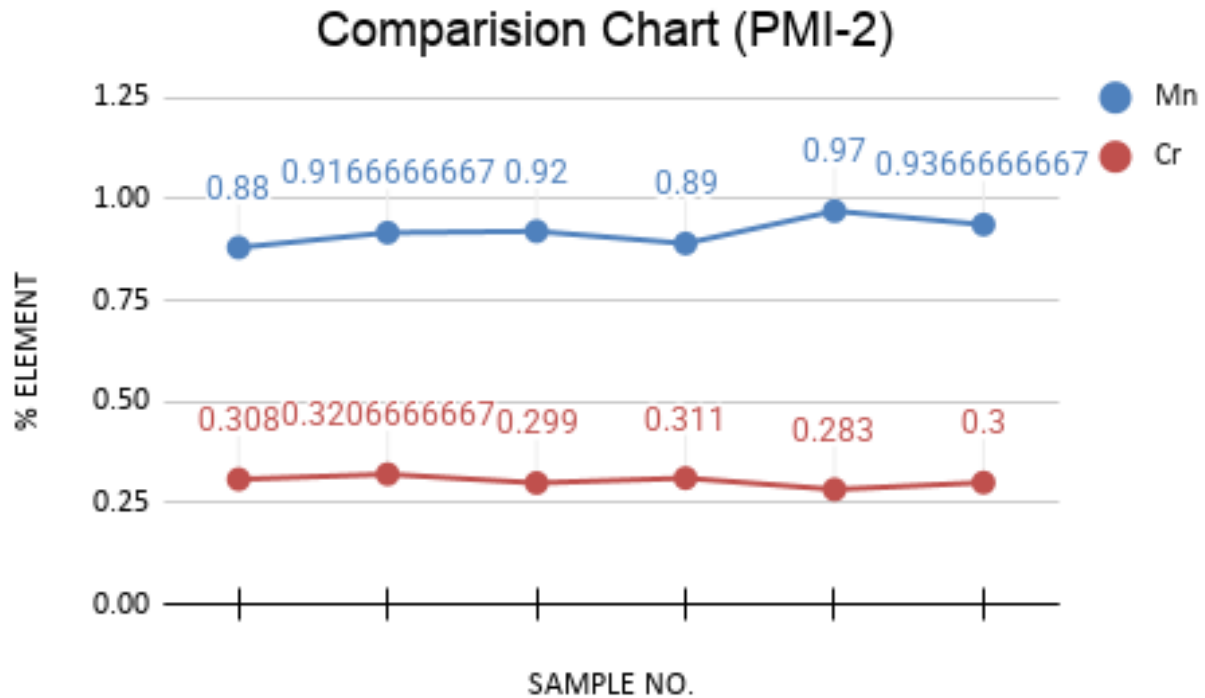


Table 2.5: Comparison of defect-location composition vs. average non-defect composition – Experiment 2

Observations

- Mn and Cr compositions show only minor variations across the four spark locations in all three samples.
- In some samples, the defect location exhibits slightly higher Mn content compared to the average of the surrounding non-defect locations.
- Since the subsurface defect is visibly present on the macro-etched surface, the higher Mn value at the defect location may indicate localized chemical heterogeneity or segregation associated with the defect region.
- Cr values remain relatively uniform and do not show any consistent enrichment at the defect location.
- The macro-etched surface may influence XRF measurements due to surface roughness and etching effects; therefore, the observed trend requires further validation.

2.8 Experiment 3: PMI Analysis after Removal of Macro-etched Layer (Surface Grinding)

The samples were ground and polished to remove the macro-etched layer and the visible defect indication, exposing a fresh metallic surface. PMI was repeated at the same radial positions (P1–P4) as in Experiment 2.

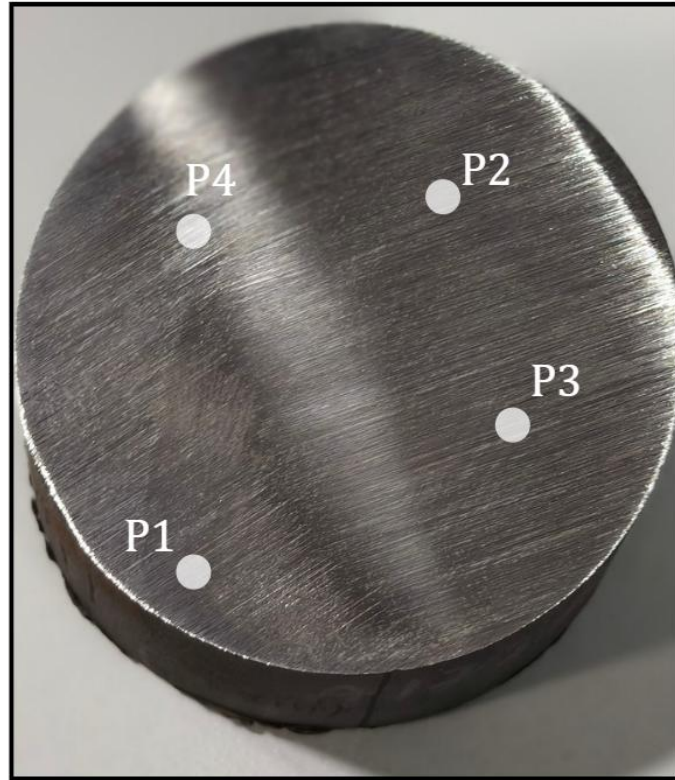


Figure 2.12: PMI spark locations – Experiment 3 (after removal of macro-etched layer)

Spark Location	Sample 1 Mn (%)	Sample 1 Cr (%)	Sample 2 Mn (%)	Sample 2 Cr (%)	Sample 3 Mn (%)	Sample 3 Cr (%)
P1 – Defect	1.000	0.305	0.960	0.270	0.950	0.304
P2 – Opposite mid-radius	0.940	0.319	0.860	0.276	0.920	0.299
P3 – Right mid-radius	0.950	0.304	0.910	0.269	0.920	0.264
P4 – Left mid-radius	0.980	0.265	0.860	0.273	0.930	0.286

Table 2.6: Mn and Cr content (%) at spark locations P1–P4, Experiment 3 (all three samples)

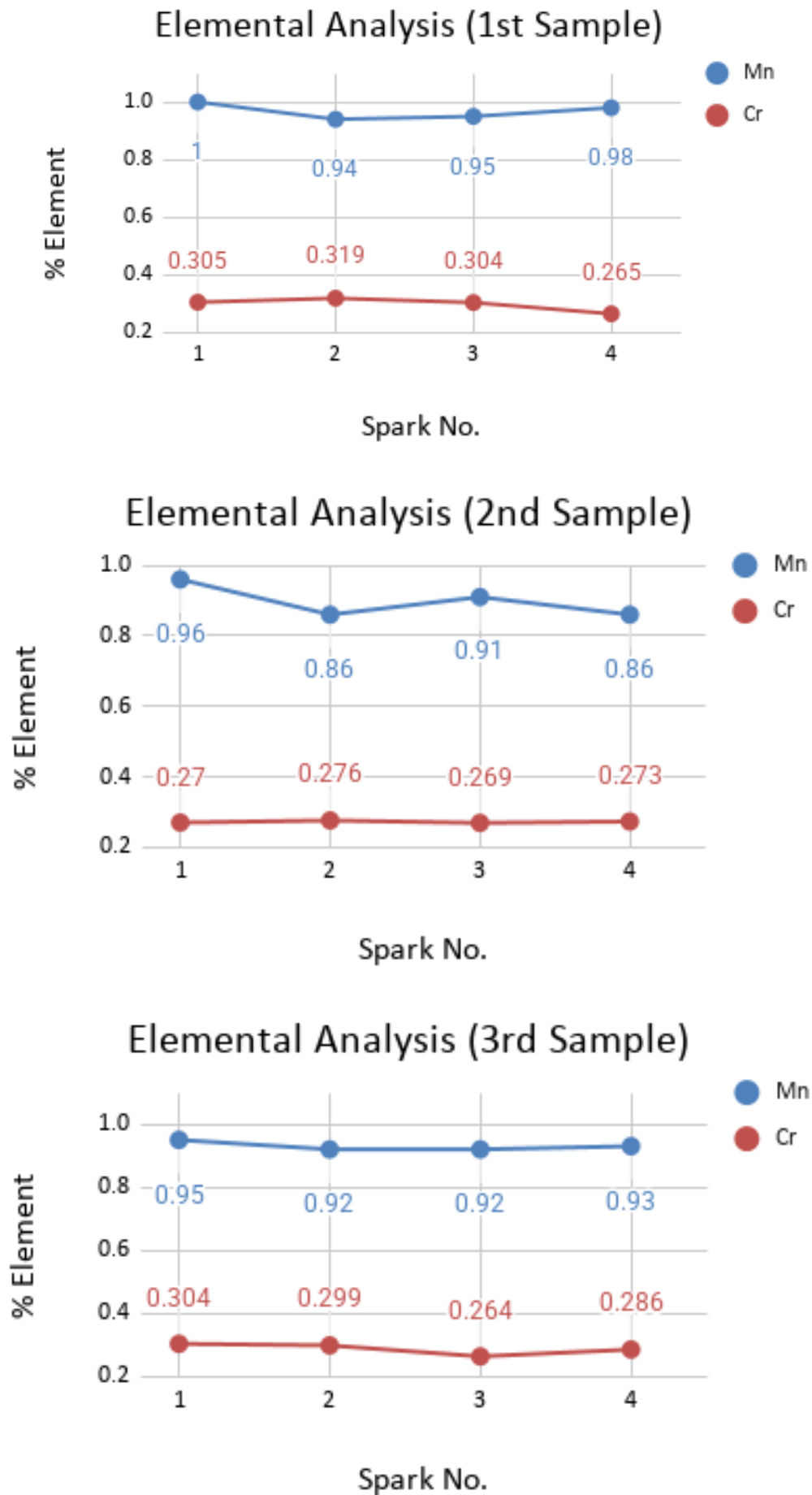


Figure 2.13: Mn and Cr variation across spark locations – Experiment 3

Sample No.	Defect (P1) Mn (%)	Non-defect Avg. Mn (%)	Defect (P1) Cr (%)	Non-defect Avg. Cr (%)
1	1.000	0.9567	0.305	0.2960
2	0.960	0.8767	0.270	0.2727
3	0.950	0.9233	0.304	0.2830

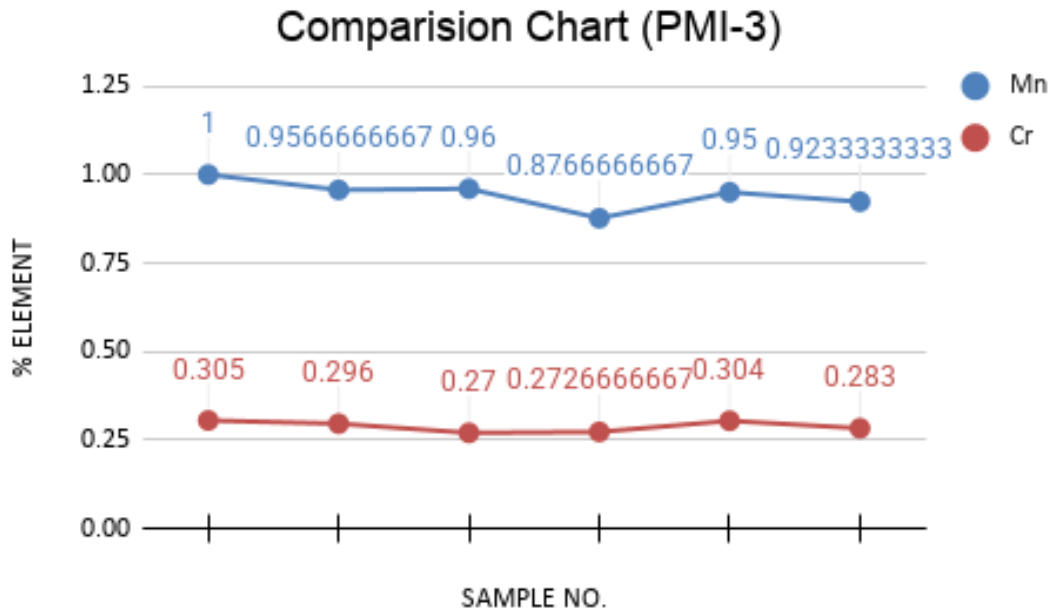


Table 2.7: Comparison of defect-location composition vs. average non-defect composition – Experiment 3

Observations

- After grinding and polishing, the macro-etched layer and visible defect indication were removed, exposing a fresh metallic surface for analysis.
- Although the defect was no longer visible, the defect location still exhibited slightly higher Mn content than the average of the surrounding non-defect locations.
- This observation suggests that localized chemical heterogeneity may still exist beneath the removed surface layer, even when the visual indication of the defect is absent.
- Cr composition remains nearly uniform across all measurement locations, indicating no significant macroscopic Cr segregation.
- The polished surface provides more consistent and reliable XRF measurements compared to the macro-etched surface.

2.9 Summary of Preliminary Investigation

The preliminary investigation on Heat No. A1726 demonstrated that surface grinding successfully exposed the underlying sound material and enabled a comparison of chemical composition beneath the defective surface. Across all three PMI experiments, Mn showed a consistent, mild enrichment at the defect / center location relative to surrounding non-defect regions, while Cr remained largely uniform. This is consistent with the expected behaviour of Mn as a positively segregating element during solidification, and supports the hypothesis that the observed surface defects are related to centerline segregation and associated sub-surface cracking originating during continuous casting, rather than to gross compositional anomalies introduced during rolling.

2.10 Need for Detailed Metallurgical Investigation

Although the preliminary investigation successfully compared the chemical composition before and after grinding, it did not provide sufficient information regarding the internal morphology, depth, or grain-level origin of the defects. Compositional analysis by PMI can indicate segregation trends but cannot resolve crack geometry, associated inclusions, or the local microstructure. Therefore, a detailed metallurgical investigation was carried out on a new rolled product belonging to Heat No. 1907, a Vanadium micro-alloyed steel, using metallographic techniques stereo and optical microscopy, controlled etching, and cross-sectional analysis to characterize the defect in greater depth. The remainder of this report (Chapters 3–7) describes this detailed investigation.

Chapter 3

Sample Details

and

Experimental

Procedure

3.1 Material Details

Parameter	Details
Heat No.	A 1907
Steel Type	Vanadium micro-alloyed steel
Product	Rolled Bar
Grade	38MnVS6
Dimension	38mm Diameter
Defect Investigated	Sub-surface Crack
No. of Samples Studied	3

Vanadium is added as a micro-alloying element to achieve grain refinement and precipitation strengthening through the formation of fine V(C,N) carbonitrides during rolling and cooling. Because such precipitates influence local mechanical response and grain growth behaviour, the metallographic investigation of Heat No. 1907 places particular emphasis on grain size and crack-microstructure interaction, in addition to defect morphology.

3.2 Sample Selection and Test Plan

Three rolled samples from Heat No. 1907 were selected for investigation. Each sample was subjected to a different, complementary line of examination as summarized below.

Metallurgical Investigation Flowchart

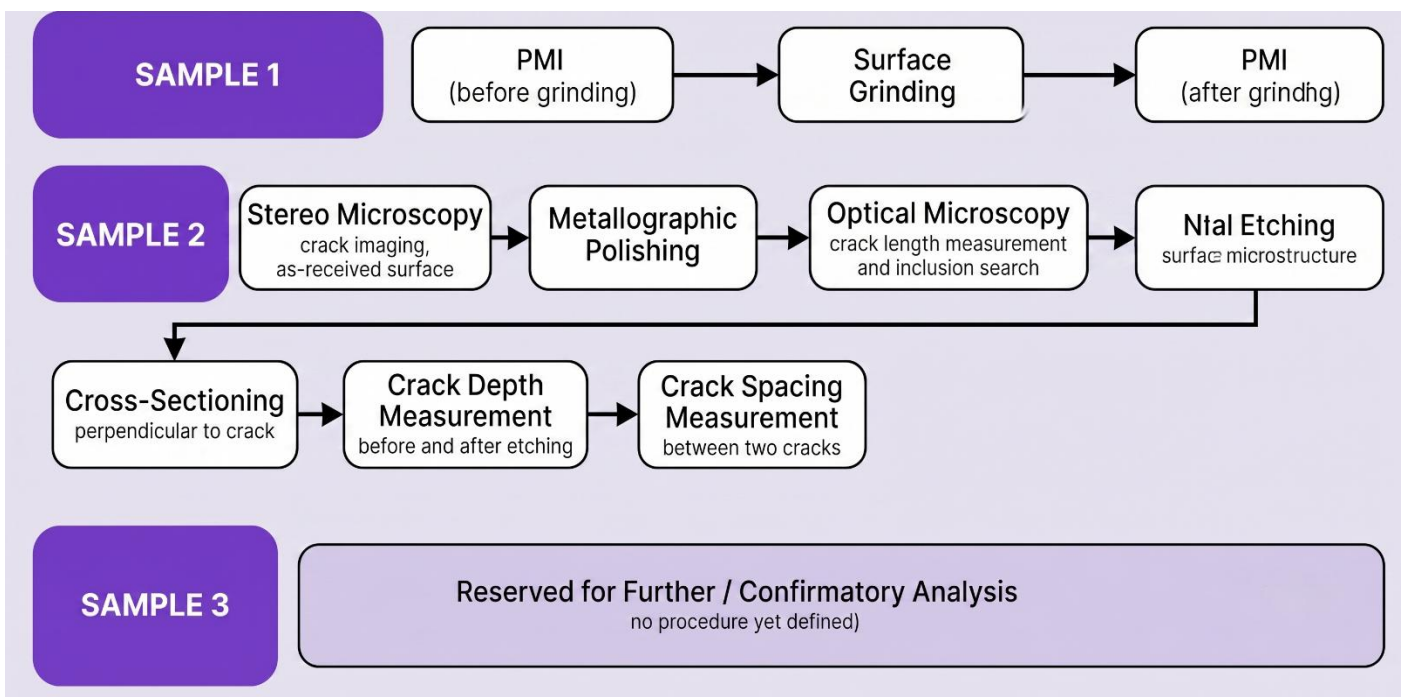


Figure 3.1: Test plan flow chart for Heat No. 1907 samples

3.3 Experimental Procedure

3.3.1 Positive Material Identification (PMI) – Sample 1

Positive Material Identification (PMI) is a rapid, non-destructive analytical technique used to verify the chemical composition of a metallic component against its specified grade, and to detect any localized compositional deviation that could be associated with a surface or sub-surface defect. In this investigation, PMI was employed as the first line of examination on Sample 1, with the specific objective of determining whether the sub-surface crack defect was associated with any chemical anomaly, such as localized segregation, contamination, or an out-of-specification alloy composition.

- PMI was performed on Sample 1 in the as-received condition using a handheld PMI analyzer (X-ray fluorescence or optical emission spectroscopy, depending on instrument availability), following the same procedure described in Section 2.4.
- Readings were taken at the defect location itself, directly over the region where the crack indication was visually observed, to capture any compositional signature specific to the defect.
- In parallel, readings were also taken at multiple surrounding non-defect locations on the same sample, spaced at representative intervals, to establish a reliable baseline composition for the bulk material.
- This dual-location testing strategy (defect vs. non-defect) was adopted specifically to isolate whether any elemental enrichment or depletion was localized to the crack region, or whether the composition was uniform throughout, which would point toward a purely mechanical or thermal origin for the defect rather than a metallurgical/chemical one.
- Key elements of interest for this micro-alloyed grade included carbon (C), manganese (Mn), sulfur (S), phosphorus (P), and vanadium (V), given their known influence on hardenability, grain refinement, and segregation behavior in 38MnVS6 steel.
- The sample surface was then ground to remove the surface layer containing the visible defect indication, exposing fresh, unaffected material directly beneath the original defect location.
- PMI was repeated at the identical measurement locations after grinding, using the same instrument settings and calibration, to obtain a directly comparable post-grinding compositional profile.
- The pre-grinding and post-grinding results were compared element-by-element, with particular attention paid to any elements showing a measurable shift between the two conditions.
- This comparison served two purposes: first, to assess whether the defect was associated with any compositional change (such as localized segregation of C, Mn, S, or P, consistent with the central segregation phenomena discussed in Section 2.2), and second, to rule out surface oxidation, decarburization, or surface contamination as confounding factors that could have biased the original as-received reading.
- The results of this PMI comparison are presented and discussed in the corresponding results section of this report.

3.3.2 Surface Grinding

Surface grinding was carried out on Sample 1 as an intermediate preparatory step between the two PMI measurement campaigns, with the specific purpose of removing the surface layer containing the visible defect indication so that the underlying, unaffected material could be independently assessed.

- Grinding was carried out using a bench grinder or belt grinder, selected based on sample geometry and the size of the region requiring material removal.

- Material was removed progressively, in a series of controlled passes, rather than in a single aggressive grinding operation, to avoid excessive local heat generation that could alter the near-surface microstructure or introduce grinding-induced thermal artifacts (such as localized tempering or untempered martensite formation) that might confound the subsequent PMI or metallographic results.
- The purpose of this grinding step was twofold. First, it served to expose sound, defect-free material beneath the visible crack indication, allowing a valid "after-grinding" PMI reading to be obtained at the same physical location as the original "before-grinding" reading. Second, it served to remove the surface oxidation layer and any decarburized zone present on the as-received surface, both of which are well known to bias PMI readings by locally altering the near-surface elemental concentration independent of the true bulk composition of the steel.
- Grinding was carried out in discrete stages, with visual inspection carried out between each pass to confirm the extent of material removal and to verify complete removal of the visible defect indication before proceeding to the repeat PMI measurement.
- Care was taken to grind only the minimum depth necessary to eliminate the surface indication, preserving as much of the original defect geometry and surrounding material as possible for any subsequent cross-referencing with the Sample 2 metallographic findings.
- The ground surface was visually and, where necessary, dimensionally documented (e.g., approximate depth of material removed) to support the interpretation of the post-grinding PMI results.

3.3.3 Metallographic Sample Preparation (Sample 2)

Metallographic sample preparation is a critical precursor to all subsequent microscopic examination, since the quality of grinding and polishing directly determines the clarity and reliability of the microstructural, crack-morphology, and inclusion observations that follow. Sample 2 was prepared specifically for detailed metallographic study of the crack, its surrounding microstructure, and its depth characteristics.

- The defect region of Sample 2 was first sectioned using a precision cut-off machine (abrasive cutting wheel with coolant) to a size convenient for standard metallographic mounting and handling, while taking particular care to preserve the crack and its immediately surrounding material intact and undamaged during the sectioning operation.
- Where necessary, the sectioned specimen was hot- or cold-mounted in a suitable mounting resin to facilitate handling during subsequent grinding and polishing operations and to support the edge retention needed for accurate near-surface and near-crack examination.
- Progressive grinding was carried out using successive grades of silicon carbide (SiC) abrasive papers, beginning with a coarse grit to remove sectioning marks and progressing sequentially through finer grits to eliminate the surface deformation introduced during the coarser grinding stages.
- Between each grinding stage, the sample was rotated approximately 90° relative to the previous grinding direction, and grinding at each stage was continued until all scratches from the preceding stage were fully removed, ensuring a progressively refined and uniformly prepared surface.
- Adequate water cooling and lubrication were maintained throughout grinding to prevent localized heating of the sample surface, which could otherwise alter the microstructure or introduce grinding artifacts in the vicinity of the crack.

- Final polishing was carried out on a rotating cloth polishing wheel using diamond or alumina polishing compound of progressively finer particle size (e.g., 6 μm , 3 μm , 1 μm), culminating in a scratch-free, mirror-like surface finish suitable for high-magnification optical examination.
- This mirror finish is essential for reliable subsequent examination, since any residual scratches, deformation, or polishing artifacts could obscure fine microstructural features, mask small non-metallic inclusions, or be misinterpreted as crack-related features during imaging.
- The prepared surface was visually inspected under low magnification prior to proceeding to imaging and etching steps, to confirm adequate polish quality and to verify that the crack remained intact and clearly visible for subsequent documentation.

3.3.4 Stereo Microscopy (All Sample)

Stereo (low-power) microscopy was employed across all samples as an initial macro-visual documentation step, providing a broad overview of the crack's surface appearance, orientation, and general morphology before proceeding to higher-magnification, sample-specific metallographic techniques.

- The sample surface was examined under a stereo microscope to image the crack in its natural, macro-etched state, allowing the general path, branching, and surface morphology of the crack to be documented across the full visible length of the defect.
- High-magnification images (relative to unaided visual inspection) of the crack were captured at multiple fields of view to comprehensively document its surface appearance, orientation relative to the rolling direction, and general morphology, including any visible branching, forking, or surface discontinuities.
- Illumination conditions (oblique lighting, ring light, or coaxial illumination, as appropriate) were adjusted as needed to maximize contrast between the crack and the surrounding sound material, ensuring clear photographic documentation.
- This stage of examination served purely as a visual and photographic documentation checkpoint, establishing a macro-scale record of the defect's appearance prior to any further sample-specific preparation; no quantitative measurements (such as crack length) or inclusion analysis were performed at this stage.
- The stereo microscopy images serve as a reference baseline against which the higher-magnification optical microscopy, cross-sectional, and etched microstructural observations (Sections 3.3.5–3.3.7) can later be correlated, allowing the overall crack morphology to be related back to its localized microstructural characteristics.

3.3.5 Nital Etching and Microstructural Examination (Sample2)

Following metallographic preparation, Sample 2 was subjected to Nital etching in order to reveal the underlying microstructure and assess its relationship to the observed crack.

- The polished surface was etched using Nital reagent, a dilute solution of nitric acid in ethanol (typically 5% concentration), which selectively attacks grain boundaries and certain microstructural constituents to reveal contrast under the optical microscope.
- Etching time was carefully controlled (typically on the order of several seconds to a few tens of seconds, depending on reagent concentration and material response) to achieve adequate contrast without over-etching, which can obscure fine structural detail, produce excessive relief, or generate etch pits that could be mistaken for genuine microstructural or defect features.

- Immediately following etching, the sample was rinsed in ethanol or water and dried to halt the etching reaction and prevent staining or residue formation on the examined surface.
- The etched surface was examined under an optical microscope at a range of magnifications to study the general microstructure in the vicinity of the crack, including grain morphology, phase distribution, and any banding or segregation-related structural features.
- Particular attention was paid to the crack path itself, assessing whether propagation occurred preferentially along grain boundaries (intergranular), through the grain interiors (transgranular), or along any segregation band or other microstructural discontinuity, since this relationship provides valuable insight into the likely mechanism of crack initiation and propagation.
- Representative micrographs were captured at multiple magnifications to document both the overall microstructural context surrounding the crack and finer local detail at the crack tip and along its flanks.

3.3.6 Cross-Sectioning and Crack Depth Measurement (Sample 2)

To fully characterize the crack in three dimensions, and specifically to determine its depth of penetration into the material a parameter that cannot be reliably assessed from a surface (plan-view) examination alone Sample 2 was sectioned transversely through the defect.

- The sample was sectioned perpendicular to the crack path using a precision cut-off machine, exposing a transverse cross-section through the defect at a location selected to be representative of the crack's overall depth and morphology.
- The cross-section was ground and polished following the same progressive grinding and polishing procedure described in Section 3.3.3, in order to obtain a scratch-free surface suitable for accurate optical measurement.
- Crack depth was first measured optically in the polished, unetched condition, using the microscope's calibrated stage and measurement software, with depth taken as the perpendicular distance from the original sample surface to the deepest visible extent of the crack tip.
- The cross-section was then etched using Nital reagent (as described in Section 3.3.5), and the crack depth was re-measured on the etched surface at the same location, in order to determine whether etching revealed any additional depth, secondary branching, or fine microstructural features (such as decarburization along the crack flanks) that were not visible in the unetched condition.
- Any discrepancy between the pre-etch and post-etch depth measurements was noted and considered in the interpretation of the true crack depth, since etching can sometimes reveal subtle secondary cracking that extends the apparent depth of the primary crack.
- During this cross-sectional examination, two distinct cracks were identified within the examined region, rather than a single continuous defect.
- The spacing (perpendicular distance) between these two cracks was measured on the polished/etched cross-section, in order to characterize their spatial distribution relative to one another and to assess whether they were likely to have originated from a common cause (e.g., a shared region of local stress concentration or segregation) or represented independent defect events.
- All measurements were repeated at a minimum of two locations along each crack where practicable, to improve the reliability and repeatability of the reported depth and spacing values.

Chapter 5

Results

5.1 Stereo Microscope Examination

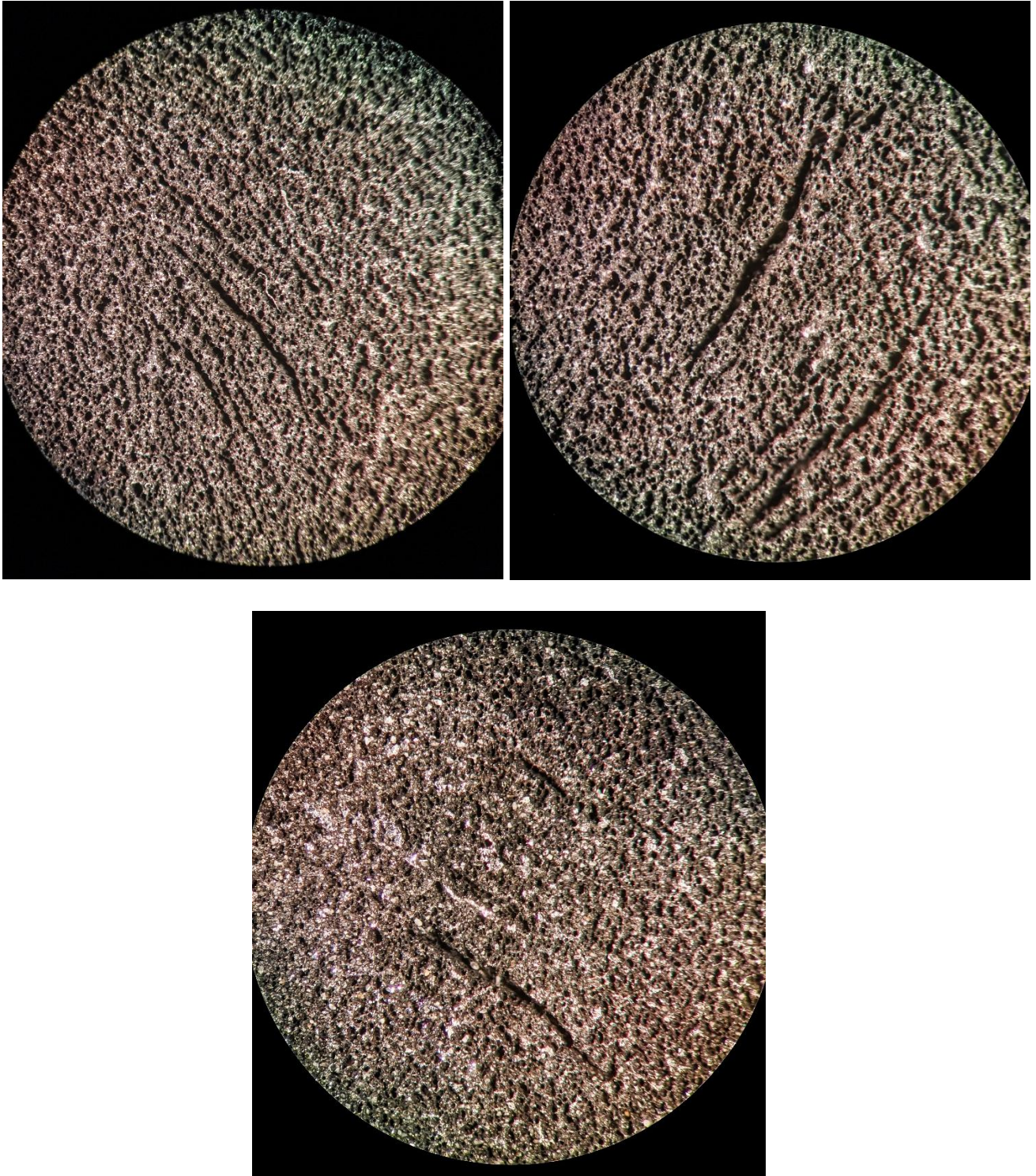


Figure 5.01: stereo Microscope Images of sample 1 ,2,3.

5.2 Sample 1 – PMI Results (Before and After Grinding)

PMI Locations

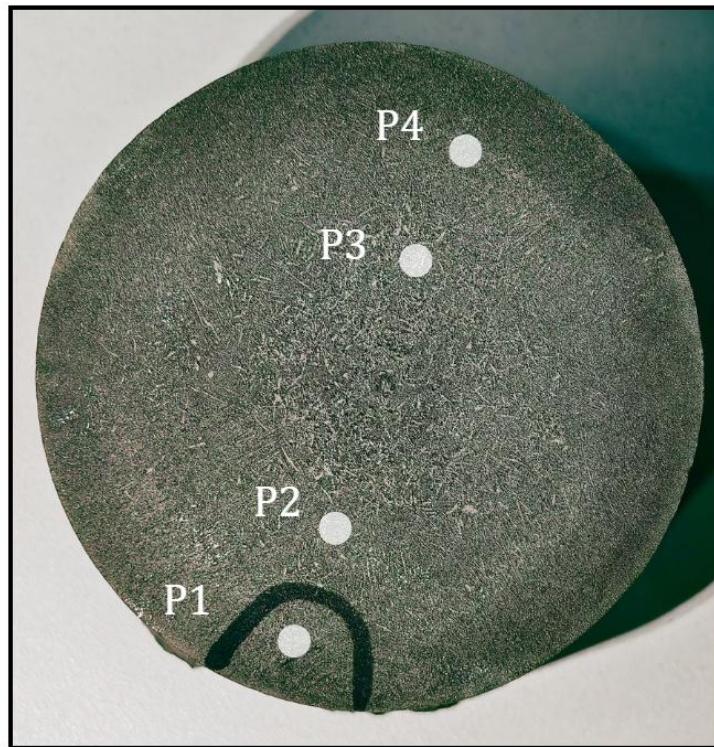


Figure 5.02: PMI Locations of Sample 1, Heat No. 1907

Element	Before Grinding Defect	Before Grinding Non-Defect	Before Grinding Non-Defect	Before Grinding Non-Defect
Mn	1.64	1.44	1.70	1.66
Cr	.183	.210	.172	.166
Ni	.101	.106	.062	-
Mo	-	.016	.012	-
Si	-	-	-	-
Cu	.162	.209	.183	-
V	.143	.163	.171	-

Table 5.1: PMI results, Sample 1, Heat No. 1907, before and after surface grinding

Element	After Grinding Defect	After Grinding Non-Defect	After Grinding Non-Defect	After Grinding Non-Defect
Mn	1.61	1.49	1.62	1.59
Cr	.146	.142	.150	.141
Ni	-	-	.110	.051
Mo	.013	.015	.014	.013
Si	-	-	-	-
Cu	.162	.172	.126	.147
V	.108	.120	.103	.113

5.3 Crack Length Measurement – Optical Micrographs (Sample 2)

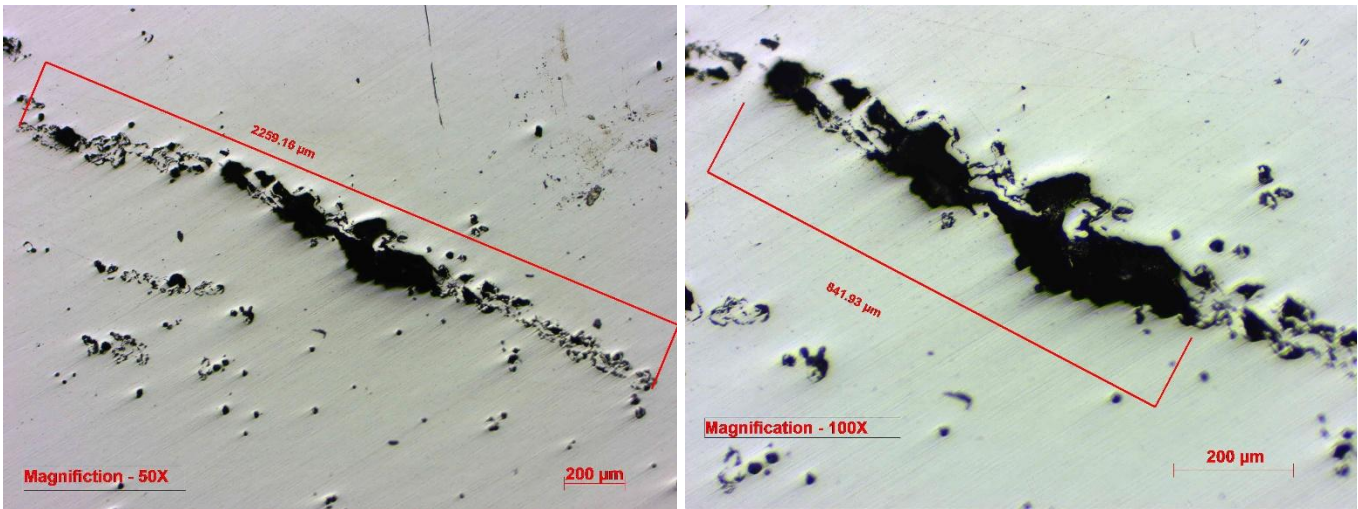


Figure 5.03: Figure (a): Crack 1 at 50X magnification, measured length = 2259.16 μm

Figure 5.3 (b): Crack 1 at 100X magnification, measured length = 841.93 μm

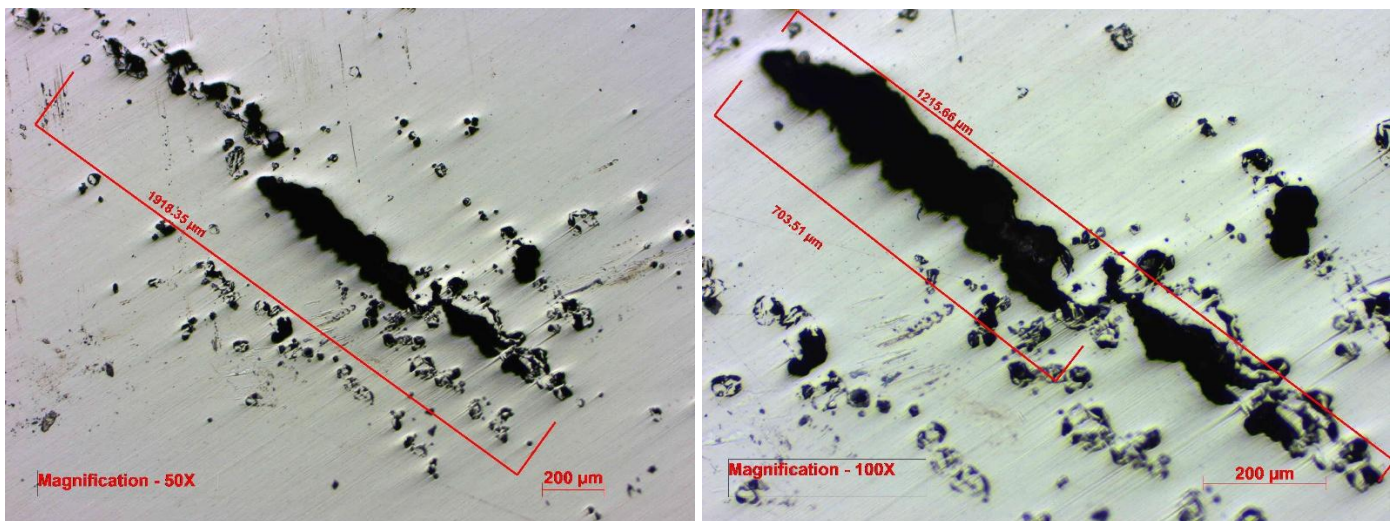


Figure 5.04(a): Crack 2 at 50X magnification, measured length = 1918.35 μm

Figure 5.4(b): Crack 2 at 100X magnification, measured lengths = 1215.66 μm and 703.51 μm

5.4 Nital-Etched Microstructure (Top Surface)

The polished surface was etched using Nital reagent to reveal the general microstructure in the vicinity of the crack. Micrographs were captured at progressively increasing magnification (50X, 100X, 200X, and 400X) to characterize the grain morphology and phase distribution at both a broad field-of-view and finer local detail.

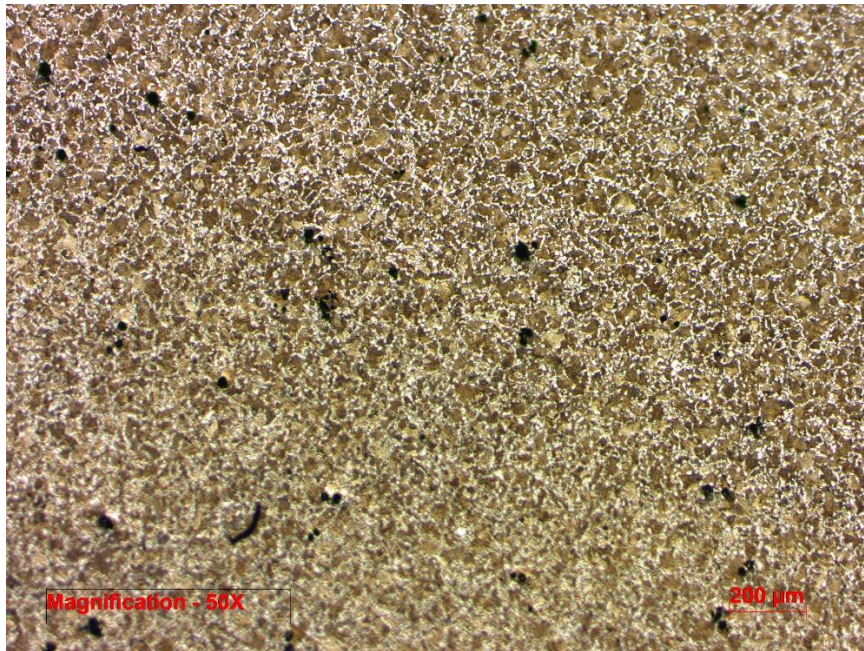


Figure 5.05: Etched microstructure at 50X magnification

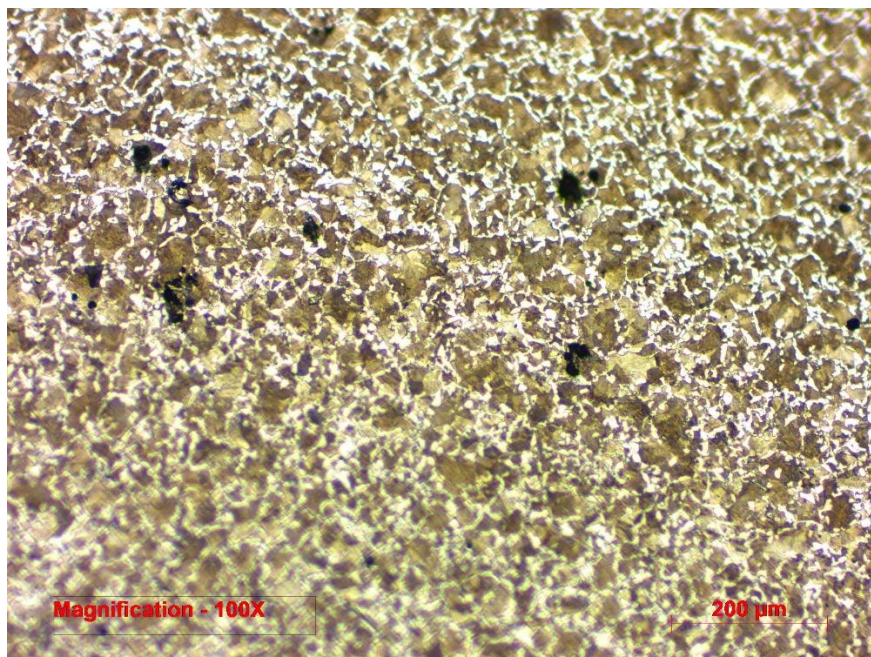


Figure 5.06: Etched microstructure at 100X magnification

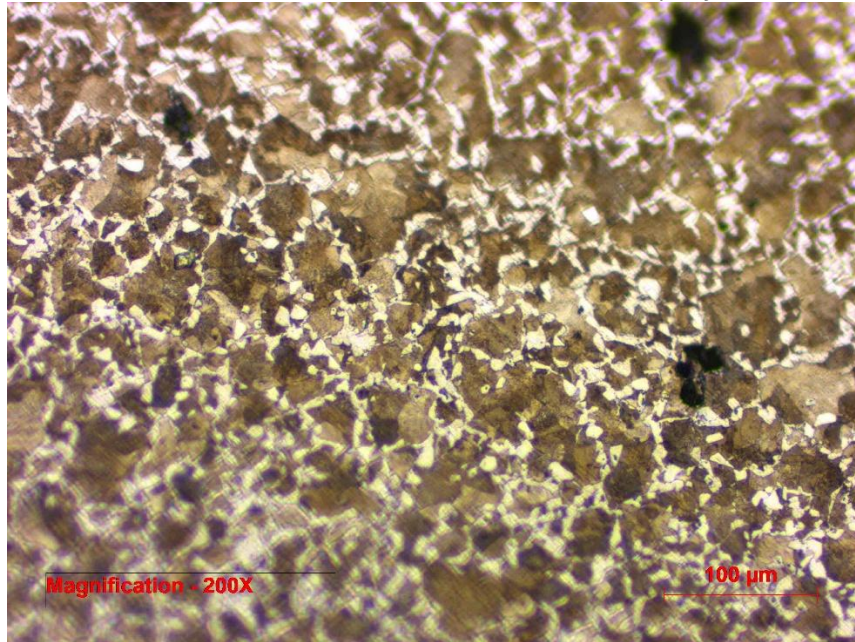


Figure 5.07: Etched microstructure at 200X magnification

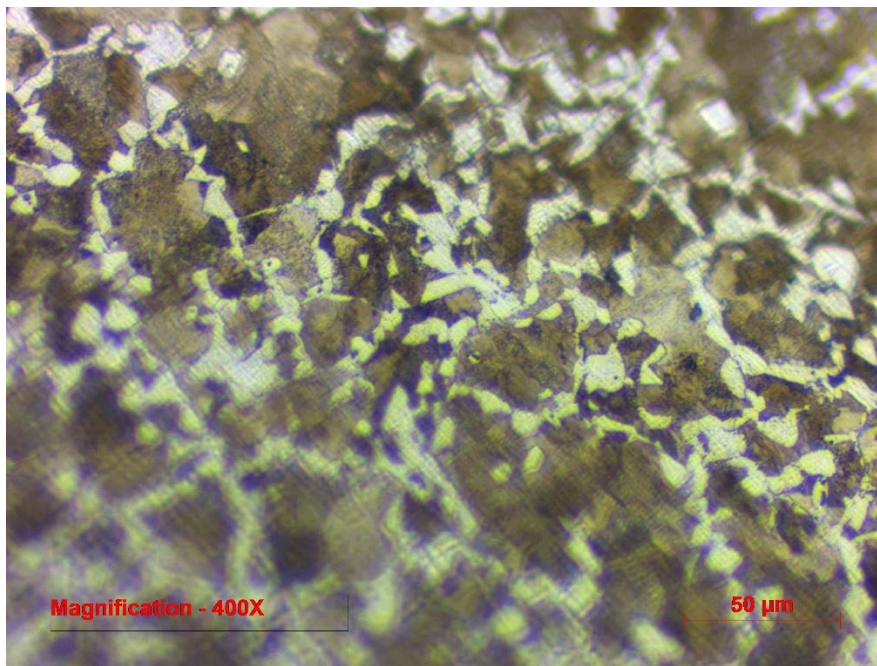


Figure 5.08: Etched microstructure at 400X magnification

The micrographs reveal a fine, relatively uniform equiaxed grain structure typical of a rolled and normalized micro-alloyed steel, with well-defined grain boundaries clearly resolved by the Nital etch. At lower magnification (50X, 100X), the overall grain distribution appears homogeneous across the field of view, with no evidence of gross banding or large-scale structural anomaly in the examined region. Scattered small dark particles, visible at all magnifications, are consistent with fine inclusions or etch-pitting rather than any continuous defect feature. At higher magnification (200X, 400X), individual grain boundaries are more clearly delineated, allowing closer observation of grain boundary morphology and the distribution of the ferrite and pearlite constituents.

5.5 Cross-Sectional Examination

5.5.1 Crack Depth

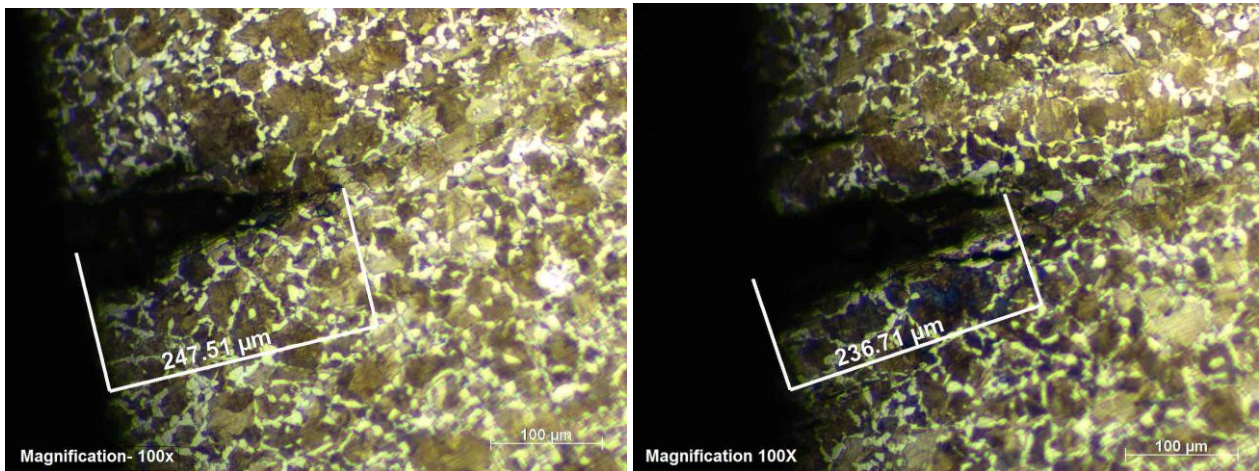


Figure 5.09: Cross-section through the crack after etching, showing crack depth measurements of 247.51 μm (crack 1) and 236.71 μm (crack 2) at 100X magnification.

To characterize the through-thickness extent of the defect, the sample was sectioned perpendicular to the crack, and the resulting cross-section was ground, polished, and etched using Nital reagent to reveal the crack path relative to the surrounding microstructure. Crack depth was measured optically at 100X magnification using the microscope's calibrated measurement tool, taken as the perpendicular distance from the sample surface to the deepest visible extent of the crack tip.

Measurements were taken at two locations along the crack, yielding depths of 247.51 μm and 236.71 μm respectively (Figure 5.6). The close agreement between these two values indicates a relatively consistent crack depth along the examined length, rather than a sharply localized or highly irregular penetration profile. In both cases, the crack is seen to terminate within the etched microstructure without any evidence of branching or secondary cracking extending significantly beyond the measured depth, and the crack path shows no clear preferential association with any single microstructural constituent visible at this magnification.

5.5.2 Crack Spacing

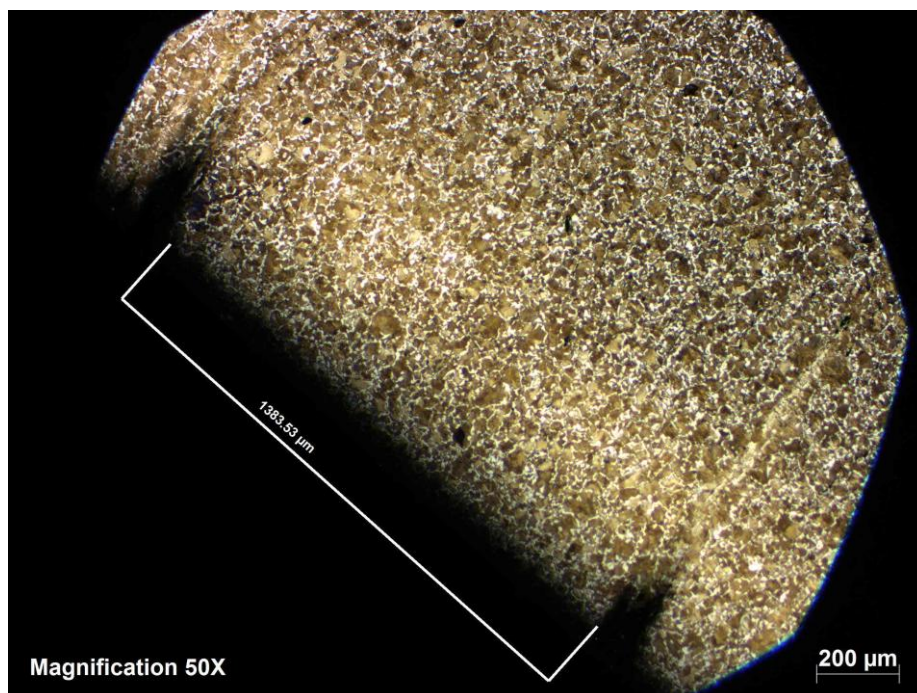


Figure 5.10: Distance between the two cracks observed in the cross-section

Chapter 6

Discussion

The findings from both stages of investigation the preliminary PMI-based study on Heat No. A1726 and the detailed metallographic study on Heat No. 1907 are discussed together below, organized by the type of evidence (chemical, morphological, microstructural) rather than by individual experiment, so as to build a single, coherent picture of the probable defect origin. Because the two heats differ in grade, product size, and nominal defect classification, agreement between them on a given point is treated as stronger evidence than either dataset would provide on its own.

6.1 Chemical Investigation (PMI)

The PMI results for Heat No. 1907, Sample 1 show the same qualitative trend already established for Heat No. A1726 in Chapter 2. Before grinding, Mn at the defect location measured 1.64%, against an average of 1.60% across the three surrounding non-defect readings an enrichment of about 0.04 percentage points. After grinding removed the visible crack indication, the comparison gave 1.61% at the defect location against a non-defect average of 1.57%, an enrichment of essentially the same magnitude. Cr showed no meaningful difference between defect and non-defect locations in either condition (0.183% vs. 0.183% before grinding; 0.146% vs. 0.144% after), mirroring the Cr uniformity already reported for Heat A1726.

This pattern a small, reproducible Mn enrichment against a flat Cr background is consistent with Mn's known tendency to segregate positively during solidification (Section 2.2.1), and its recurrence in an independent heat and grade strengthens the case that it reflects a real metallurgical trend rather than a one-off reading. Vanadium is a more ambiguous case: it showed a mild depletion at the defect location before grinding (0.143% vs. a non-defect average of 0.167%), but this narrowed substantially after grinding (0.108% vs. 0.112%) unlike Mn, whose enrichment held at a similar magnitude in both conditions. That difference in behaviour suggests the pre-grinding V reading may carry a surface-related contribution rather than reflecting genuine bulk segregation, so it is reported here for completeness rather than as a firm conclusion. Cu, Ni, Mo, and Si could not be reliably compared: several readings for these elements are missing across the two PMI campaigns (Table 5.1), and where data does exist, Cu shows an inconsistent trend lower than the non-defect average before grinding, higher after. This is treated as a data limitation rather than evidence either way.

Taken together, the PMI evidence indicates that compositional segregation is present and is a contributing factor in Heat 1907, in the same manner it was for Heat A1726 but the magnitude involved is mild rather than gross, so it is best read as having produced a locally weakened, solute-enriched zone rather than being sufficient by itself to explain the crack. Sections 6.3–6.5 address how that chemical predisposition relates to the crack actually observed.

6.2 Surface Condition and Effect of Grinding

Grinding served the same purpose and produced the same outcome in both heats. In Heat A1726 (Section 2.8, Experiment 3), grinding removed the macro-etched layer and the visible defect indication, exposing a fresh surface for PMI yet the defect-location Mn value remained above the local non-defect average in all three samples (1.000% vs. 0.957% in Sample 1; 0.960% vs. 0.877% in Sample 2; 0.950% vs. 0.923% in Sample 3). The same pattern repeats in Heat 1907: grinding Sample 1 removed the visible crack indication, and the post-grind reading at the defect location (1.61% Mn) still sat above the surrounding non-defect average (1.57%), essentially unchanged from the 1.64% vs. 1.60% comparison obtained before grinding.

This is a useful negative result. If the elevated Mn reading had been a surface artefact oxide scale, a decarburized layer, macro-etchant residue, or surface contamination grinding away the surface layer should have removed or substantially reduced the differential. Instead, it persists to the depth removed by grinding in both heats, which places the origin of the Mn enrichment in the bulk material established during solidification, rather than in anything introduced at or after the rolling or finishing stage. This also validates surface grinding, as used here, as a reliable way to obtain a genuine sound-material PMI baseline rather than one confounded by residual surface effects.

6.3 Crack Morphology (Stereo Microscopic Observations)

Two physically distinct cracks were documented and measured on the surface of Sample 2. Crack 1 measured 2259.16 μm at 50X magnification, with a further reading of 841.93 μm at 100X the higher-magnification field of view captures only part of the full crack, so this second figure is a local segment rather than a discrepancy. Crack 2 provides a useful check on the measurement method itself: its 50X length (1918.35 μm) is reproduced almost exactly by the sum of its two 100X segment lengths (1215.66 μm + 703.51 μm = 1919.17 μm , a difference of under 1 μm), which gives confidence that the optical length measurements in Section 5.3 are internally consistent.

Both cracks are of the same order of magnitude in length (roughly 1.9–2.3 mm), pointing to a common formation mechanism and comparable severity rather than one being a minor feature next to a dominant one. Read alongside the depth results in Section 6.5, both cracks are roughly eight to nine times longer than they are deep (Crack 1: 2259.16 μm long against 247.51 μm deep; Crack 2: 1918.35 μm long against 236.71 μm deep). A crack this much longer than it is deep is not the signature expected of a fresh, isolated mechanical event after rolling, which would tend toward a more equiaxed shape it is instead consistent with a small, pre-existing casting-stage discontinuity that has been drawn out preferentially along the rolling direction during hot rolling, while its through-thickness extent is comparatively little affected. This length-to-depth relationship is one of the stronger pieces of evidence for a casting, rather than rolling or in-service, origin.

6.4 Inclusion Analysis

No non-metallic inclusions were found in the material surrounding the crack in Sample 2. This is a meaningful negative result rather than an absence of data: Section 5.4 notes that the general microstructure does contain scattered small dark particles, interpreted there as fine inclusions or etch-pitting typical of any commercially produced steel but critically, none of these particles are concentrated at, or aligned along, the crack itself. Had the crack been associated with a refractory particle or an inclusion stringer (Section 2.2.5), a chain or cluster of non-metallic particles would be expected along its path. Their absence makes a refractory-defect or exogenous-inclusion origin unlikely, and points the probable cause toward the thermal- and segregation-related mechanisms discussed above rather than an entrapment-related one.

6.5 Microstructural Observations

The Nital-etched top-surface micrographs (Section 5.4) show a fine, relatively uniform equiaxed grain structure typical of a rolled and normalized micro-alloyed steel, with ferrite and pearlite constituents clearly resolved at 200X and 400X. No gross banding or large-scale structural anomaly is visible in the examined region, which together with the absence of inclusions noted in Section 6.4 indicates that the surrounding matrix is otherwise soundly processed and that the defect is a localized, discrete feature rather than a symptom of a pervasive matrix problem.

The cross-sectional examination (Section 5.5.1) adds the through-thickness picture. Crack depth was measured at two locations, giving 247.51 μm and 236.71 μm close enough (within about 5%) to indicate a fairly consistent depth of penetration along the examined length rather than a sharply localized or irregular one. At both locations the crack terminates cleanly within the etched microstructure, with no branching or secondary cracking beyond the measured depth, and its path shows no clear preferential association with any single microstructural constituent at the magnification examined. This is more consistent with a crack driven by a broadly weakened local zone of the kind implied by the mild Mn enrichment in Section 6.1 than one dictated by a specific microstructural feature such as a grain-boundary film or banded phase.

Crack 1 and Crack 2 also lie close to one another within the same region of the cross-section (Section 5.5.2, Figure 5.7). Given that they are additionally similar in both length (2259.16 μm vs. 1918.35 μm) and depth (247.51 μm vs. 236.71 μm), the more likely explanation is that they share a common local origin two adjacent manifestations of the same underlying segregation/stress condition rather than representing two independent defect events at unrelated locations in the billet.

6.7 Probable Origin of the Defect

Bringing the evidence in Sections 6.1–6.5 together, the crack in Heat No. 1907, Sample 2 is most consistent with a continuous-casting-related sub-surface discontinuity, initiated in a mildly Mn-enriched

(locally segregated) zone during solidification, that was subsequently exposed and substantially elongated in the rolling direction during hot rolling.

Each line of evidence contributes a specific part of this picture. The PMI results (Section 6.1) establish that a mild, reproducible Mn enrichment is present at the defect location in both heats examined, and that this enrichment survives grinding (Section 6.2) and therefore originates below the surface rather than from surface contamination. The absence of non-metallic inclusions along the crack (Section 6.4) rules out a refractory or exogenous-particle origin, leaving a thermal- or segregation-related mechanism as the more probable cause. The surrounding matrix is otherwise fine-grained and free of gross banding (Section 6.5), so the defect is a localized event rather than a symptom of a heat-wide casting or rolling problem. And the crack morphology itself (Section 6.3) millimeter-scale in length but only a few hundred microns deep, with two similarly-sized cracks in close proximity is the signature expected of a small casting-stage flaw drawn out along the rolling direction, rather than one that initiated fresh during or after rolling.

This conclusion mirrors, independently, the one reached for Heat A1726 in Chapter 2, where centerline-segregation-associated surface defects were similarly linked to a consistent, mild Mn enrichment trend across three separate PMI experiments. That two different grades, product sizes, and nominal defect types converge on the same chemical signature and the same general mechanism strengthens confidence in the conclusion for both.

This picture should be treated as well-supported but provisional until the grain size comparison in Section 6.6 is completed. A measurably coarser grain size at the crack tip than in the bulk would directly confirm localized thermal/casting influence at that location and complete the case made here; a uniform fine grain size would not contradict the conclusion above, but would mean it rests on the chemical and morphological evidence alone, without additional microstructural confirmation.

Chapter 7

Conclusions

- Macrostructural defects observed in rolled steel products of Heat No. A1726 (Grade 080A47) were identified as Centerline Segregation and Sub-Surface Cracks, consistent with known continuous-casting-related discontinuities.
- PMI analysis across three complementary experiments (linear, macro-etched surface, and post-grinding) consistently showed a mild Mn enrichment at the defect/center location relative to surrounding non-defect regions, while Cr remained essentially uniform supporting a segregation-related origin rather than a gross compositional anomaly.
- Surface grinding effectively removed the visible defect indication in both heats and exposed sound material for reliable compositional comparison. Critically, the Mn enrichment persisted after grinding in both heats (1.000% vs. 0.957% in Heat A1726, Sample 1; 1.61% vs. 1.57% in Heat 1907, Sample 1), indicating the enrichment originates in the bulk material rather than from a surface artefact such as oxidation or contamination.
- Detailed metallographic investigation of Heat No. 1907 (Vanadium micro-alloyed steel, Grade 38MnVS6) combined PMI, stereo/optical microscopy, Nital and Picral etching, and cross-sectional analysis to characterize the surface crack in Sample 2 in terms of length, depth, spacing, associated microstructure, and grain size.
- Two distinct cracks were identified on Sample 2, measuring 2259.16 μm and 1918.35 μm in length (50X) and 247.51 μm and 236.71 μm in depth respectively each roughly eight to nine times longer than it is deep, a proportion consistent with a small casting-stage discontinuity subsequently elongated along the rolling direction during hot rolling.
- No non-metallic inclusions were found in association with either crack, indicating the defect is not refractory- or inclusion-driven.
- The microstructure surrounding the crack was fine-grained, uniformly equiaxed, and free of gross banding, indicating the defect is a localized event rather than a symptom of a heat-wide casting or rolling problem.
- Overall, the combined chemical (PMI) and metallographic evidence points toward the observed surface/sub-surface cracks in both heats originating during continuous casting most likely at a mildly Mn-segregated zone and being subsequently exposed and elongated during hot rolling.

Chapter 8

Supplementary

Investigation

of Additional

Samples

8.1 Introduction to Supplementary Investigation

Following the detailed metallurgical investigation of Heat No. 1907, additional steel samples were provided for analysis. Given that these samples lacked accompanying heat records or pre-assigned material grades, a revised investigative protocol was implemented. The primary objective of this supplementary study was twofold: first, to establish the base material specification through rapid identification techniques, and second, to validate the previously established defect characterization methodology specifically the synergy between stereo microscopy and PMI on material in its "as-received" state. This approach allowed for a direct assessment of defect morphology and localized elemental distribution without the influence of prior surface grinding, providing a critical cross-reference to the data obtained from the ground surfaces of the primary heats.

8.2 Experimental Methodology

The investigative protocol for these five supplementary samples was designed to characterize material properties and defect signatures without relying on pre-existing heat documentation. The procedure followed four sequential, logical stages:

- **As-Received Stereo Microscopy:** Without any surface preparation, four of the samples were examined under a stereo microscope to document the crack morphology. This step was essential for capturing the native state of the defect including its path, branching, and surface orientation without the risk of introducing artifacts that could occur during grinding or polishing.
- **Material Identification via Spectroscopy:** Since heat records were unavailable for these samples, optical emission spectroscopy was employed as the initial step to determine the chemical grade. This process provided the necessary baseline to confirm whether the steel conformed to expected industrial standards and ensured subsequent compositional data could be accurately interpreted.
- **Targeted PMI Analysis:** Once the material grade was verified through spectroscopy, PMI (XRF) was conducted directly on both the defective and non-defective regions of the four samples mentioned above. This analysis provided a compositional signature of the defect in its native, unground state, which served as a crucial reference for evaluating the presence of elemental segregation prior to any material removal.
- **Metallographic Microstructural Analysis:** For the fifth sample, a detailed metallographic investigation was performed. This involved cross-sectioning through the defect region, followed by progressive grinding and polishing to a mirror finish. The sample was then etched to reveal the grain structure and crack-tip characteristics under high-magnification optical microscopy. This final stage provided critical data on the through-thickness extent of the cracks and the influence of the local microstructure on defect propagation.

8.3 Results and Observations

The rapid investigation of the five supplementary samples provided a comprehensive overview of the crack characteristics, ranging from initial as-received visual screening to detailed microstructural assessment.

8.3.1 As-Received Stereo Microscopy (Samples 1–4): Each of the four samples designated for rapid screening was examined in the as-received condition using stereo microscopy. This provided essential documentation of the defect morphology, path, and surface distribution prior to any surface modification. The high-magnification images in Figure 8.1 through 8.4 demonstrate the crack path and surface distribution of the defects across the varying sample geometries.

As-Received Stereo Microscopy (Samples 1):

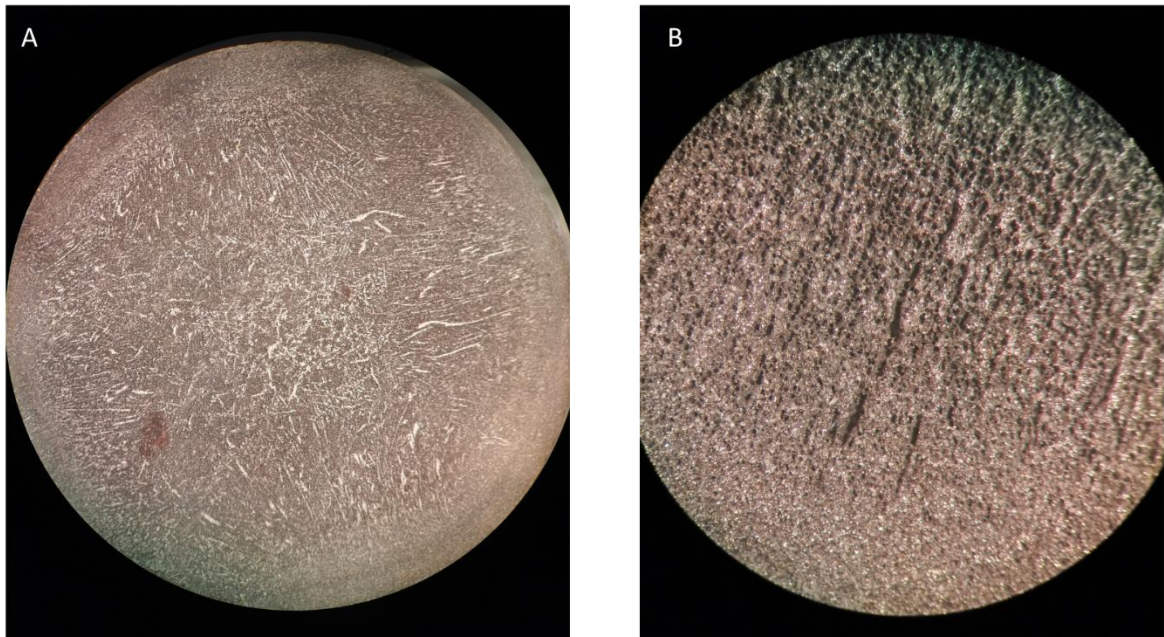


Figure 8.1: As-received stereo microscopic images of Sample 1. (A) Full surface view showing the general macrostructure and crack location. (B) Magnified view of the single surface crack.

As-Received Stereo Microscopy (Samples 2):

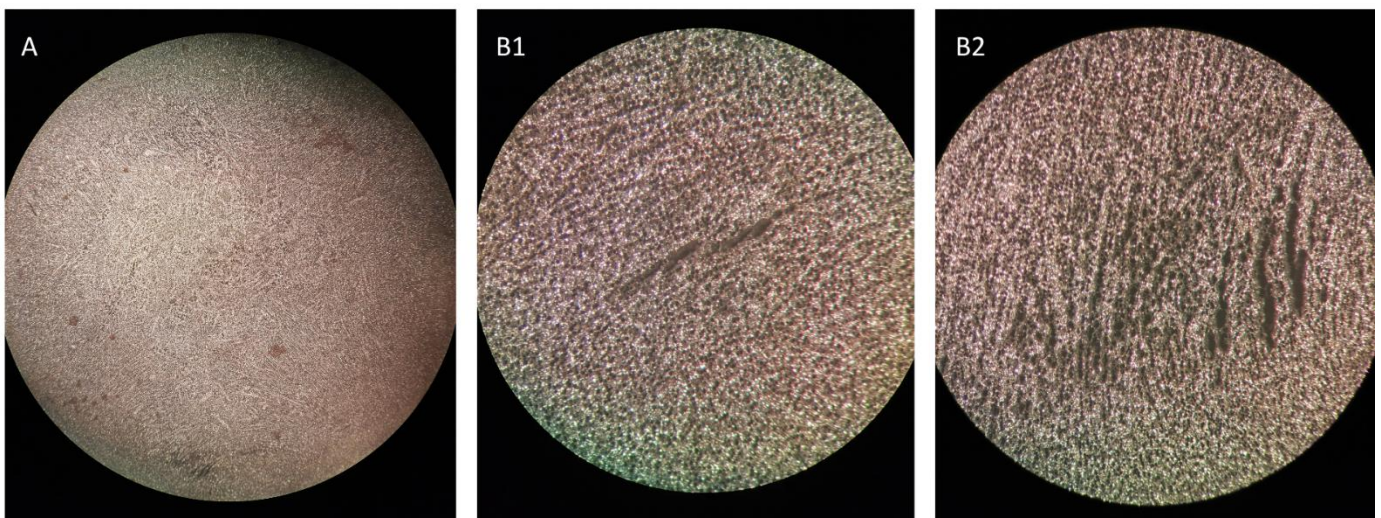


Figure 8.2: As-received stereo microscopic images of Sample 2. (A) Full surface view showing the general macrostructure. (B1) and (B2) Magnified views detailing the presence of multiple **subsurface cracks**.

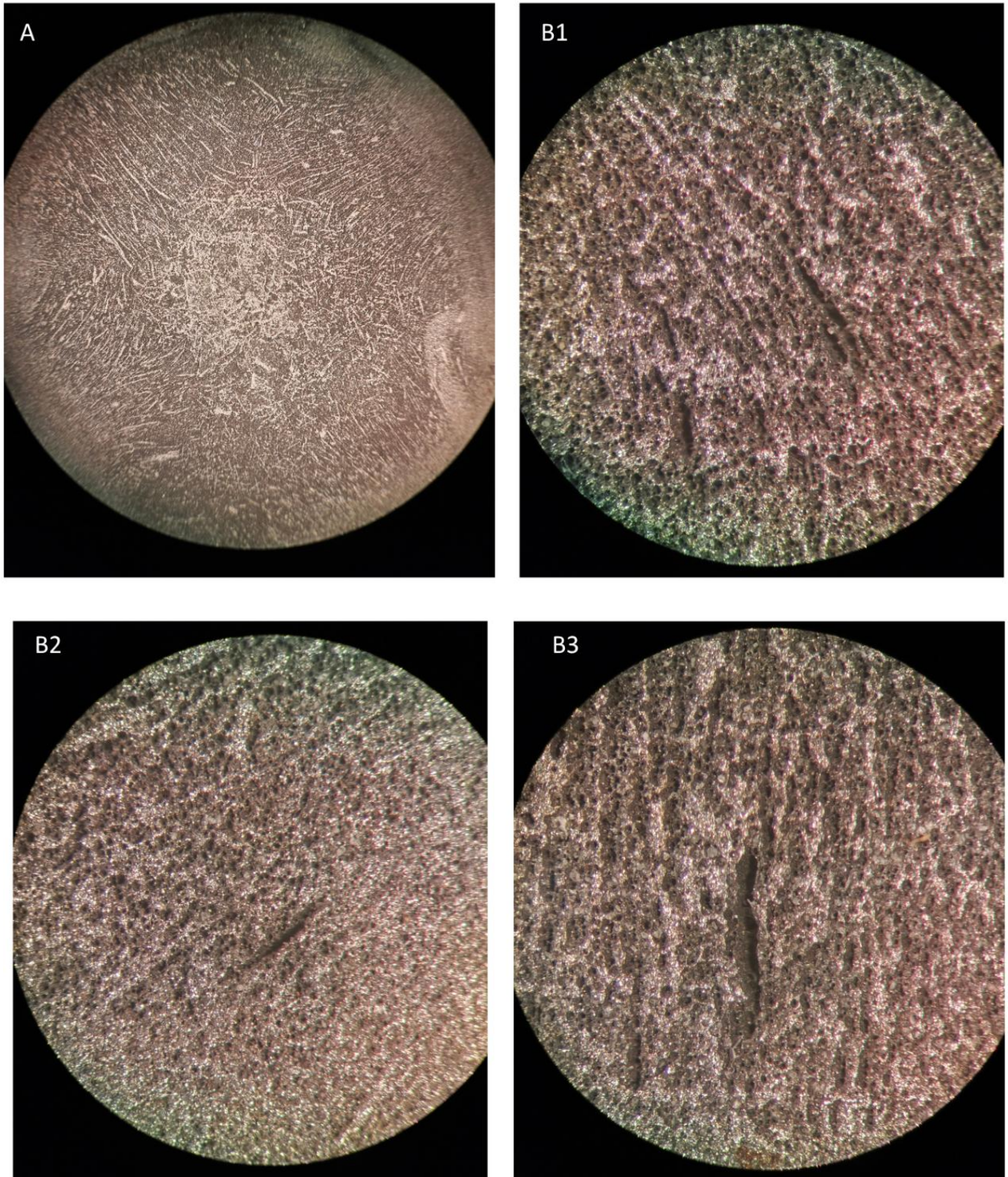
As-Received Stereo Microscopy (Samples 3):

Figure 8.3: As-received stereo microscopic images of Sample 2. (A) Full surface view showing the general macrostructure. (B1), (B2), (B3) Magnified views detailing the presence of multiple **subsurface cracks**.

As-Received Stereo Microscopy (Samples 4):

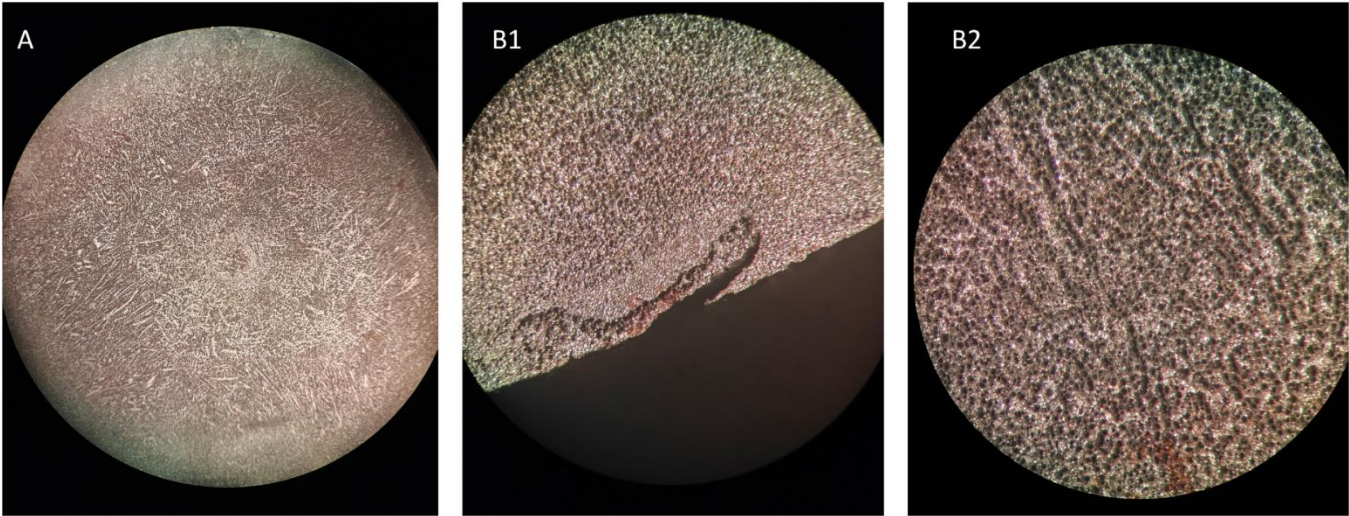
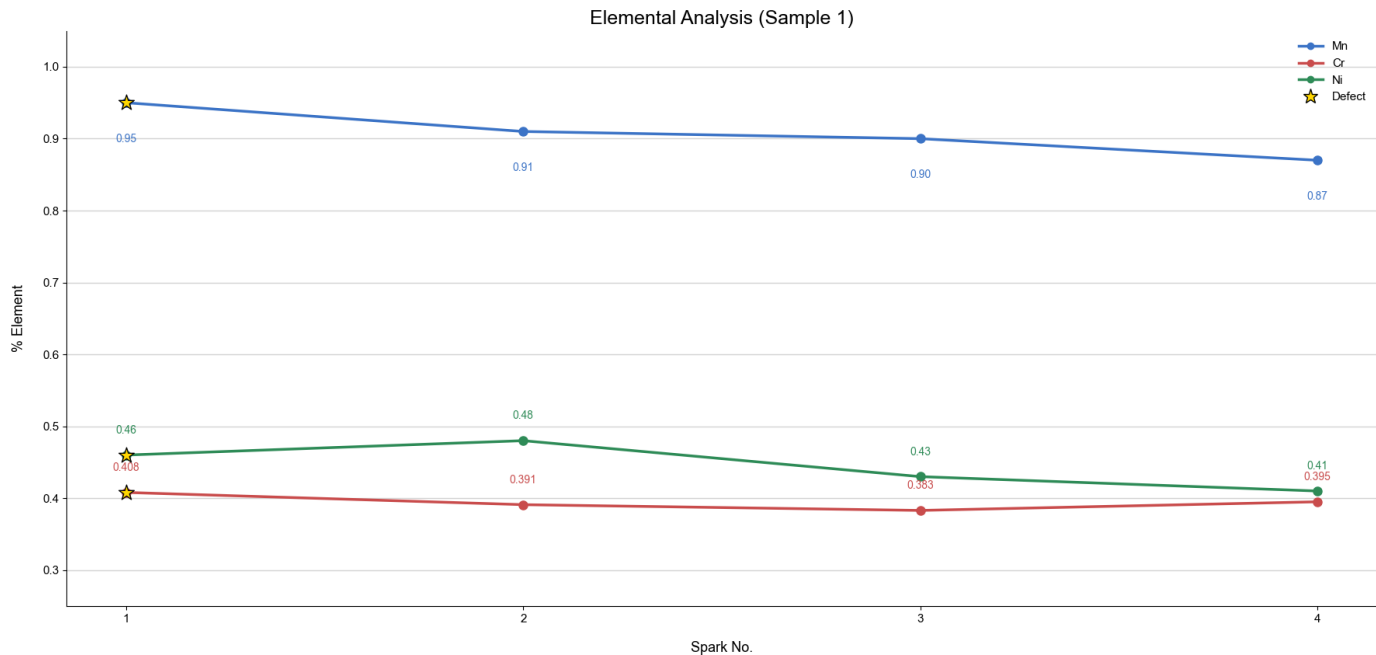


Figure 8.4: As-received stereo microscopic images of Sample 2. (A) Full surface view showing the general macrostructure. (B1) is a Seam and (B2) is Magnified views detailing the presence of multiple **subsurface cracks**.

8.3.2 Chemical Analysis (Spectroscopy & PMI, Samples 1–4): Following grade identification by optical emission spectroscopy, Positive Material Identification (PMI) analysis was performed on both the defective and non-defective regions of Samples 1–4. The elemental composition obtained from these measurements is presented graphically in **Figure 8.5**, illustrating the variation in manganese (Mn) and chromium (Cr) contents across the different spark locations. These results establish the baseline chemical composition of each sample and help assess whether the localized manganese enrichment observed in the primary heats is consistently present in these additional material batches.



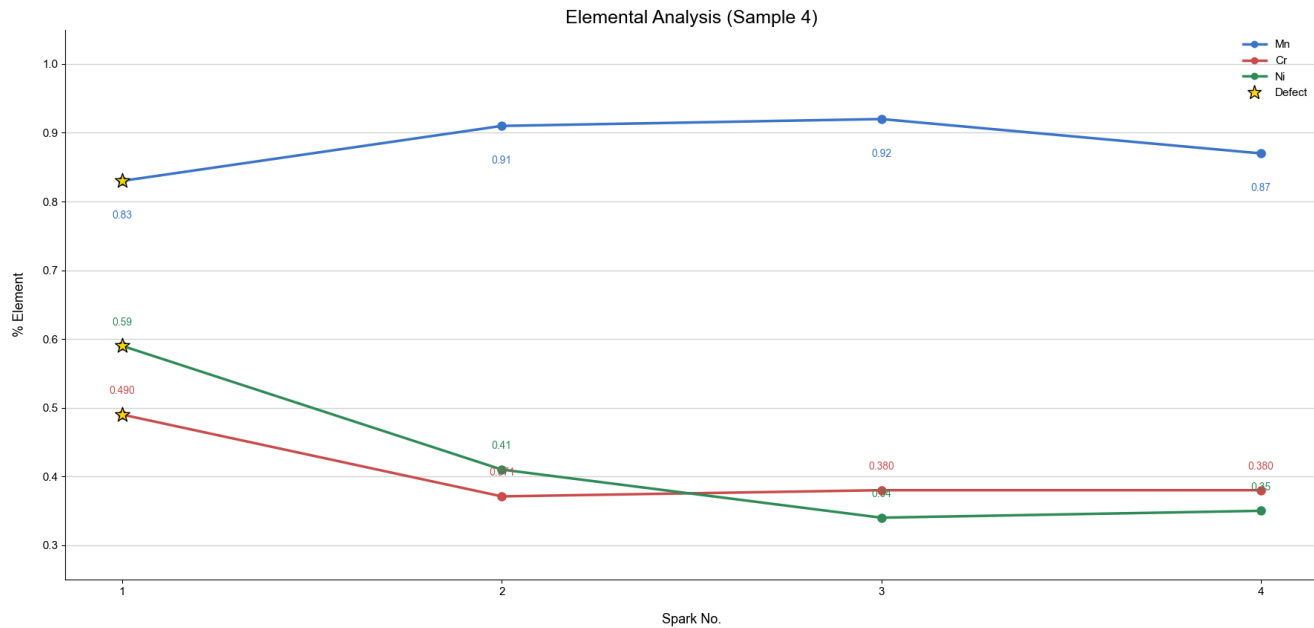
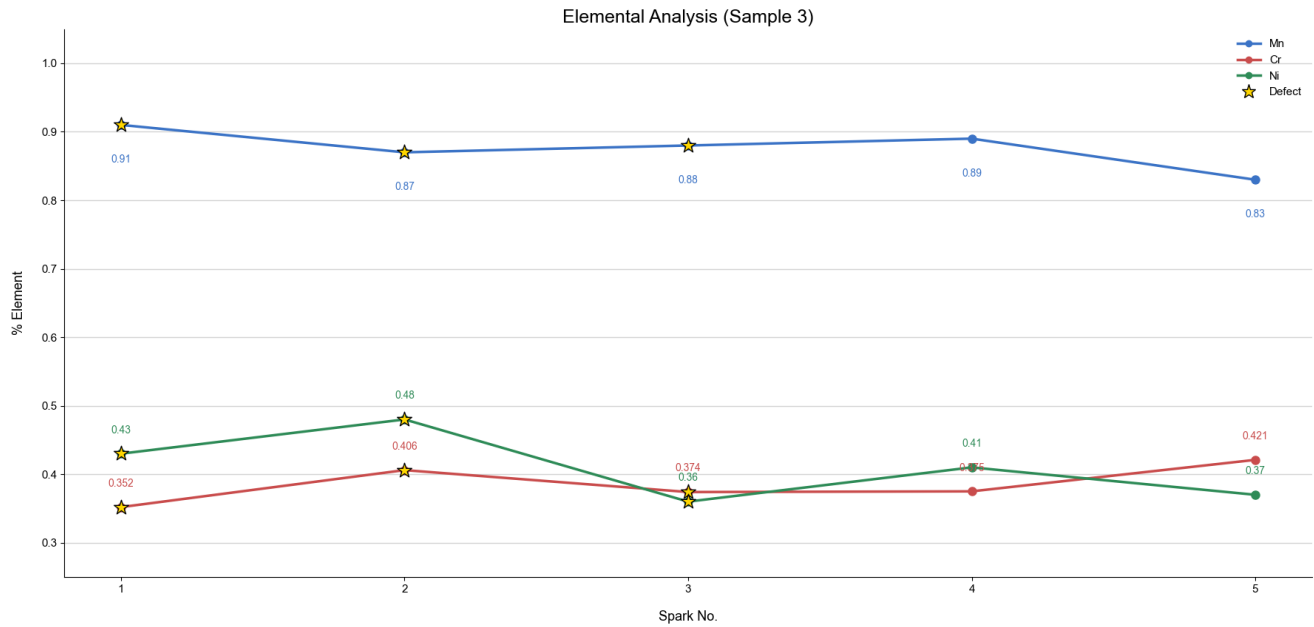
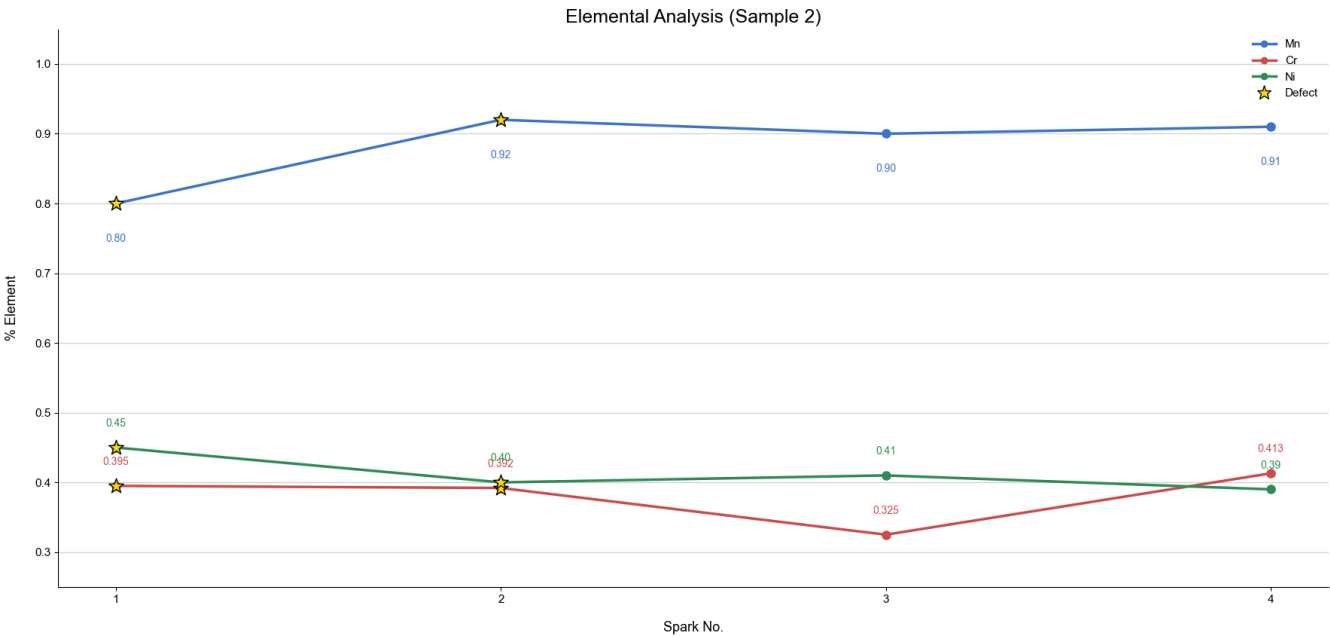


Figure 8.5: Variation of manganese (Mn) and chromium (Cr) concentrations measured by PMI at defective and non-defective locations for Samples 1–4. Defect locations are highlighted to facilitate comparison of localized compositional variations within each sample.

8.3.3 Metallographic Cross-Sectional Analysis (Sample 5): The fifth sample underwent detailed metallographic study to characterize the crack morphology and its interaction with the steel matrix. The specimen was cross-sectioned through the crack, ground, and polished to a mirror finish.

- **Crack Morphology (Unetched Condition):** Initial examination was performed on the unetched, polished surface using oblique lighting focused at the sample edge. This technique provided excellent contrast, clearly revealing the crack as a distinct, dark discontinuity against the reflective metallic background. As shown in **Figure 8.5**, this observation confirms the crack exists independently of any chemical etching process, ruling out the possibility that the crack is an etch-induced artifact.

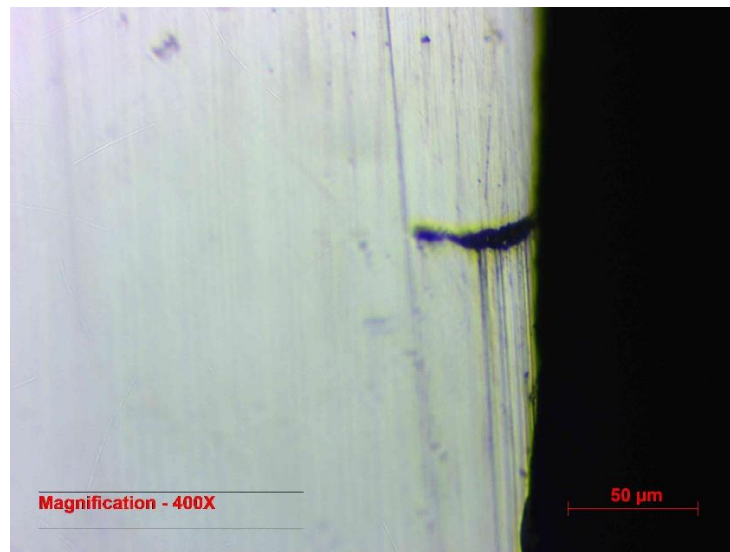


Figure 8.6: Metallographic analysis of Sample 5. Unetched cross-section under oblique lighting, clearly revealing the subsurface crack as a dark discontinuity against the metallic background.

- **Microstructural Examination (Etched Condition):** Following the unetched documentation, the sample was etched with Nital reagent to reveal the underlying ferrite-pearlite grain structure. High-magnification images at 200X and 400X (**Figures 8.6 A and B**) show a fine, equiaxed grain structure. The crack path was observed in relation to the surrounding grains, providing critical data on the through-thickness extent of the defect and the local matrix response.

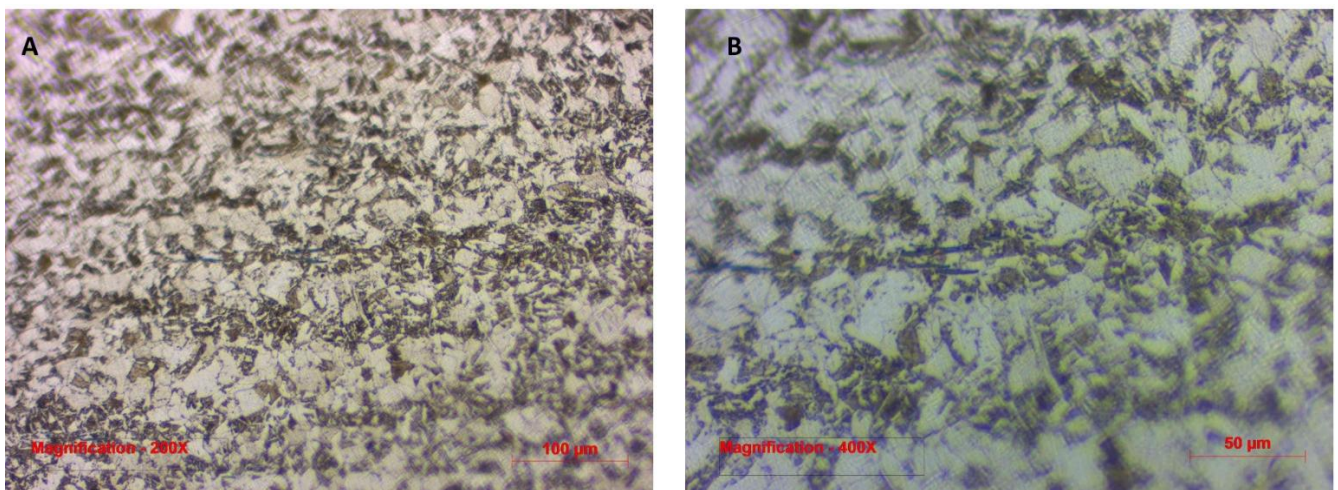


Figure 8.7 (A) Nital-etched microstructure at 200X magnification, showing the fine, equiaxed grain structure and the crack path. (B) High-magnification (400X) view of the equiaxed grain structure.